33

Alternating-Current Circuits

CHAPTER OUTLINE

33.1 AC Sources

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* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ33.1 Answer (c). When a power source, AC or DC, is first connected to a *RL* combination, the presence of the inductor impedes the buildup of a current in the circuit. The value of the current starts at zero and increases as the back emf induced across the inductor decreases in magnitude.

OQ33.2 **(i)** Answer (e). Inductive reactance, $X_L = \omega L$, doubles when the frequency doubles, so the rms current is halved $\left(I_{\text{rms}} = \Delta V_{\text{rms}}/X_L\right)$.

(ii) Answer (b). Capacitive reactance, $X_C = 1/\omega C$, is cut in half when frequency doubles, so the rms current doubles $\left(I_{\text{rms}} = \Delta V_{\text{rms}}/X_C\right)$.

(iii) Answer (d). The resistance remains unchanged $\left(I_{\text{rms}} = \Delta V_{\text{rms}}/R\right)$.

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OQ33.3 Answer (a). The voltage across the capacitor is proportional to the stored charge. This charge, and hence the voltage ΔV_C, is a maximum when the current has zero value and is in the process of reversing direction after having been flowing in one direction for a half cycle. Thus, the voltage across the capacitor lags behind the current by 90°. The capacitive reactance, $X_C = 1/\omega C$, decreases as frequency increases, causing the impedance to decrease and the current to increase.

OQ33.4 **(i)** Answer (d). $\Delta V_{\text{avg}} = \dfrac{\Delta V_{\max}}{2}$.

(ii) Answer (c). The average of the squared voltage is $\left([\Delta V]^2\right)_{\text{avg}} = \dfrac{(\Delta V_{\max})^2}{2}$. Then its square root is $\Delta V_{\text{rms}} = \dfrac{\Delta V_{\max}}{\sqrt{2}}$.

OQ33.5 Answer (d). If the voltage is a maximum when the current is zero, the voltage is either leading or lagging the current by 90° (or a quarter cycle) in phase. Thus, the element could be *either* an inductor or a capacitor. It could not be a resistor since the voltage across a resistor is always in phase with the current. If the current and voltage were out of phase by 180°, one would be a maximum in one direction when the other was a maximum value in the opposite direction.

OQ33.6 **(i)** Answer (e). The voltage varies between +170 V and –170 V.

(ii) Answer (c). The average of a sine waveform is zero.

(iii) Answer (b). $\Delta V_{\text{rms}} = \Delta V_{\max}/\sqrt{2} = 170\text{ V}/\sqrt{2} = 120\text{ V}$.

OQ33.7 **(i)** Answer (a). We have:

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$= \left\{(20.0\ \Omega)^2 + \left[2\pi(500\text{ Hz})\left(120\times10^{-3}\text{ H}\right) - \frac{1}{2\pi(500\text{ Hz})\left(0.750\times10^{-6}\text{ F}\right)}\right]\right\}^{1/2}$$

$$Z = 51.5\ \Omega$$

and $I_{\text{rms}} = \Delta V_{\text{rms}}/Z = (120\text{ V})/(51.5\ \Omega) = 2.33\text{ A}$.

(ii) Answer (b). At the resonance frequency, $X_L = X_C$ and the impedance is $Z = \sqrt{R^2 + (X_L - X_C)^2} = R$. Thus, the rms current is

$$I_{\text{rms}} = \Delta V_{\text{rms}}/Z = (120\text{ V})/(20.0\ \Omega) = 6.00\text{ A}$$

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OQ33.8 Answer (e) is *false*. In an *RLC* circuit, the instantaneous voltages Δv_R, Δv_L, and Δv_C (across the resistance, inductance, and capacitance respectively) are not in phase with each other. The instantaneous voltage Δv_R is in phase with the current, Δv_L leads the current by 90°, while Δv_C lags behind the current by 90°. The instantaneous values of these three voltages do add algebraically to give the instantaneous voltages across the *RLC* combination, but the rms voltages across these components do not add algebraically. The rms voltages across the three components must be added as vectors (or phasors) to obtain the correct rms voltage across the *RLC* combination.

OQ33.9 Answer (c). At resonance the inductive reactance and capacitive reactance cancel out.

OQ33.10 Answer (c). At resonance the inductive reactance and capacitive reactance add to zero: $\phi = \tan^{-1}[(X_L - X_C)/R] = 0$.

OQ33.11 The ranking is (a) > (d) > (b) > (c) > (e). At the resonance frequency $f_0 = 1\,000$ Hz both X_L and X_C are equal: call their mutual value X. A high-Q value means the resonance has a small width, so X_L and X_C are also much larger than R at f_0. Inductive reactance X_L is proportional to frequency, and capacitive reactance X_C is inversely proportional to frequency. In terms of X, the choices have the values: (a) $f = f_0/2$, so $X_C = 2X$. (b) $f = 3f_0/2$, so $X_C = 2X/3$. (c) $f = f_0/2$, so $X_L = X/2$. (d) $f = 3f_0/2$, so $X_L = 3X/2$. (e) R is independent of frequency, and R is less than X. Thus, we have (a) $2X$ > (d) $3X/2$ > (b) $2X/3$ > (c) $X/2$ > (e) less than X.

OQ33.12 Answer (e). The battery produces a constant current in the primary coil, which generates a constant flux through the secondary coil. With no change in flux through the secondary coil, there is no induced voltage across the secondary coil.

OQ33.13 Answer (c). AC ammeters and voltmeters read rms values. With an oscilloscope you can read a maximum voltage, or test whether the average is zero.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ33.1 (a) The Q factor determines the selectivity of the radio receiver. For example, a receiver with a very low Q factor will respond to a wide range of frequencies and might pick up several adjacent radio stations at the same time. To discriminate between 102.5 MHz and 102.7 MHz requires a high-Q circuit.

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(b) Typically, lowering the resistance in the circuit is the way to get a higher quality resonance.

CQ33.2 (a) The second letter in each word stands for the circuit element.

For an inductor L, the emf $\mathcal{E}$ leads the current I—thus ELI. For a capacitor C, the current leads the voltage across the device. In a circuit in which the capacitive reactance is larger than the inductive reactance, the current leads the source emf—thus ICE.

(b) CIVIL – in a capacitor C the current (I) leads voltage (represented by V), voltage leads current in an inductor L.

CQ33.3 The voltages are not added in a scalar form, but in a vector form, as shown in the phasor diagrams throughout the chapter. Kirchhoff's loop rule is true at any instant, but the voltages across different circuit elements are not simultaneously at their maximum values. Do not forget that an inductor can induce an emf in itself and that the voltage across it is 90° *ahead* of the current in the circuit in phase.

CQ33.4 (a) In an *RLC* series circuit, the phase angle depends on the source frequency. At very low frequency, the capacitor dominates the impedance and the phase angle is near –90°. At very high frequencies, the inductor dominates the impedance and the phase angle is near –90°.

(b) When the inductive reactance equals the capacitive reactance, the frequency is the resonance frequency; the phase angle is zero.

CQ33.5 In 1881, an assassin shot President James Garfield. The bullet was lost in his body. Alexander Graham Bell invented the metal detector in an effort to save the President's life. The coil is preserved in the Smithsonian Institution. The detector was thrown off by metal springs in Garfield's mattress, a new invention itself. Surgeons went hunting for the bullet in the wrong place. Garfield died.

CQ33.6 (a) The person is doing work at a rate of $P = Fv\cos\theta$.

(b) Compare the previous equation to $P = \Delta V_{\text{rms}} I_{\text{rms}} \cos\phi$. One can consider the emf as the "force" that moves the charges through the circuit, and the current as the "speed" of the moving charges. The $\cos\theta$ factor measures the effectiveness of the cause in producing the effect. Theta is an angle in real space for the vacuum cleaner and phi is the analogous angle of phase difference between the emf and the current in the circuit.

CQ33.7 The circuit can be considered an *RLC* series circuit.

(a) Yes. The circuit is in resonance because the inductive reactance and capcitive reactance are equal, so the total impedance $Z = R$.

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(b) Total power output by the emf $P_{\text{emf}} = I^2R_{\text{total}}$, where $R_{\text{total}} = 10\ \Omega$ (source resistance) + 10 Ω (load resistance) = 20 Ω. Power delivered to the load $P_{\text{load}} = I^2R_L$, where $R_L = 10\ \Omega$. Fraction of average power delivered to the load to average power delivered by the source of emf:

$$\frac{P_{\text{load}}}{P_{\text{emf}}} = \frac{I^2R_{\text{load}}}{I^2R_{\text{total}}} = \frac{R_{\text{L}}}{R_{\text{total}}} = \frac{R_{\text{L}}}{R_{\text{source}} + R_{\text{L}}} = \frac{10\ \Omega}{20\ \Omega} = 0.5$$

(c) The resistance of the load could be increased to make a greater *fraction* of the emf's power go to the load. Then the emf would put out a lot less power and less power would reach the load.

CQ33.8 No. A voltage is only induced in the secondary coil if the flux through the core changes in time. No changing current, no changing flux, no induced voltage.

CQ33.9 (a) The capacitive reactance is proportional to the inverse of the frequency. At higher and higher frequencies, the capacitive reactance approaches zero, making a capacitor behave like a wire.

(b) As the frequency goes to zero, the capacitive reactance approaches infinity—the resistance of an open circuit.

CQ33.10 The ratio of turns indicates the ratio of voltages: $N_1/N_2 = \Delta V_1/\Delta V_2$, where $\Delta V_2 = 120$ V; therefore, $\Delta V_1 = 12$ kV. In its intended use, the transformer takes in energy by electric transmission at 12 kV and puts out nearly the same energy by electric transmission at 120 V. With the small generator putting energy into the secondary side of the transformer at 120 V, the primary side has 12 kV induced across it. It is deadly dangerous for the repairman.

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 33.1 AC Sources

Section 33.2 Resistors in an AC Circuit

P33.1 (a) The rms voltage across the resistor is given by

$$\Delta V_{R,\text{rms}} = I_{\text{rms}}R = (8.00\ \text{A})(12.0\ \Omega) = \boxed{96.0\ \text{V}}$$

(b) From Equation 33.5,

$$\Delta V_{R,\text{max}} = \sqrt{2}\left(\Delta V_{R,\text{rms}}\right) = \sqrt{2}\,(96.0\ \text{V}) = \boxed{136\ \text{V}}$$

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(c) From Equation 33.4,

$$I_{\max} = \sqrt{2} I_{\text{rms}} = \sqrt{2}\,(8.00\ \text{A}) = \boxed{11.3\ \text{A}}$$

(d) $P_{\text{avg}} = I_{\text{rms}}^2 R = (8.00\ \text{A})^2 (12.0\ \Omega) = \boxed{768\ \text{W}}$

P33.2 The rms voltage is

$$\Delta V_{\text{rms}} = \frac{170\ \text{V}}{\sqrt{2}} = 120\ \text{V}$$

(a) $P = \dfrac{(\Delta V_{\text{rms}})^2}{R} \quad \rightarrow \quad R = \dfrac{(120\ \text{V})^2}{75.0\ \text{W}} = \boxed{193\ \Omega}$

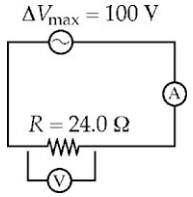
(b) $R = \dfrac{(120\ \text{V})^2}{100\ \text{W}} = \boxed{144\ \Omega}$

P33.3 Each meter reads the rms value.

(a) $I_{\text{rms}} = \dfrac{\Delta V_{\text{rms}}}{R} = \dfrac{\left(\Delta V_{\max}/\sqrt{2}\right)}{R} = \dfrac{\left(100\ \text{V}/\sqrt{2}\right)}{24.0\ \Omega} = \boxed{2.95\ \text{A}}$

(b) The voltage across the resistor is the same as that across the power supply:

$$\Delta V_{\text{rms}} = \frac{\Delta V_{\max}}{\sqrt{2}} = \frac{100\ \text{V}}{\sqrt{2}} = \boxed{70.7\ \text{V}}$$



ANS. FIG. P33.3

P33.4 (a) We compute the peak voltage from the rms voltage:

$$\Delta V_{R,\max} = \sqrt{2}\left(\Delta V_{R,\text{rms}}\right) = \sqrt{2}\,(120\ \text{V}) = \boxed{170\ \text{V}}$$

(b) From the definition of power,

$$P_{\text{avg}} = I_{\text{rms}}^2 R = \frac{\Delta V_{\text{rms}}^2}{R}$$

Solving for the resistance,

$$R = \frac{\Delta V_{\text{rms}}^2}{P_{\text{avg}}} = \frac{(120\ \text{V})^2}{60.0\ \text{W}} = \boxed{2.40 \times 10^2\ \Omega}$$

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(c) Because $P_{\text{avg}} = \dfrac{(\Delta V_{\text{rms}})^2}{R} \quad \rightarrow \quad R = \dfrac{(\Delta V_{\text{rms}})^2}{P_{\text{avg}}}$, a 100-W bulb has less resistance than a 60.0-W bulb.

P33.5 The current as a function of time is $i = \dfrac{\Delta v}{R} = \left(\dfrac{\Delta V_{\max}}{R}\right)\sin \omega t$. Given the value of t, we want to identify a point with

$$0.600\,\frac{\Delta V_{\max}}{R} = \frac{\Delta V_{\max}}{R}\sin(\omega t)$$

or $\omega t = \sin^{-1} 0.600$

To find the lowest frequency we choose the smallest angle satisfying this relation:

$$(0.007\ 00\ \text{s})\omega = \sin^{-1}(0.600) = 0.644\ \text{rad}$$

Thus, $\omega = 91.9\ \text{rad/s} = 2\pi f$ so $\boxed{f = 14.6\ \text{Hz}}$

P33.6 (a) From Equation 33.3, $\Delta v_R = \Delta V_{\max} \sin \omega t$. To find the angular frequency, we write

$$\Delta v_R = 0.250(\Delta V_{\max})$$

so $\sin \omega t = 0.250$

or $\omega t = \sin^{-1}(0.250)$

The smallest angle for which this is true is $\omega t = 0.253$ rad . Thus, if $t = 0.010\ 0$ s,

$$\omega = \frac{0.253\ \text{rad}}{0.010\ 0\ \text{s}} = \boxed{25.3\ \text{rad/s}}$$

(b) The second time when $\Delta v_R = 0.250(\Delta V_{\max})$, $\omega t = \sin^{-1}(0.250)$ again. For this occurrence, $\omega t = \pi - 0.253\ \text{rad} = 2.89\ \text{rad}$ (to understand why this is true, recall the identity $\sin(\pi - \theta) = \sin\theta$ from trigonometry). Thus,

$$t = \frac{2.89\ \text{rad}}{25.3\ \text{rad/s}} = \boxed{0.114\ \text{s}}$$

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P33.7 We are given $\Delta V_{\max} = 15.0$ V and $R_{\text{total}} = 8.20\ \Omega + 10.4\ \Omega = 18.6\ \Omega$. The maximum current in the circuit is

$$I_{\max} = \frac{\Delta V_{\max}}{R_{\text{total}}} = \frac{15.0\text{ V}}{18.6\ \Omega} = 0.806\text{ A} = \sqrt{2} I_{\text{rms}}$$

And the power delivered to the speakers is

$$P_{\text{speaker}} = I_{\text{rms}}^2 R_{\text{speaker}} = \left(\frac{0.806\text{ A}}{\sqrt{2}}\right)^2 (10.4\ \Omega) = \boxed{3.38\text{ W}}$$

P33.8 All lamps are connected in parallel with the voltage source, so $\Delta V_{\text{rms}} = 120$ V for each lamp. Also, the current is $I_{\text{rms}} = P_{\text{avg}}/\Delta V_{\text{rms}}$ and the resistance is $R = \Delta V_{\text{rms}}/I_{\text{rms}}$.

(a) For the 150-W bulbs,

$$I_{\text{rms}} = \frac{150\text{ W}}{120\text{ V}} = 1.25\text{ A}$$

For the 100-W bulb,

$$I_{\text{rms}} = \frac{100\text{ W}}{120\text{ V}} = 0.833\text{ A}$$

The rms current in each 150-W bulb is 1.25 A. The rms current in the 100-W bulb is 0.833 A.

(b) The resistance in bulbs 1 and 2 is

$$R_1 = R_2 = \frac{120\text{ V}}{1.25\text{ A}} = \boxed{96.0\ \Omega}$$

and the resistance in bulb 3 is

$$R_3 = \frac{120\text{ V}}{0.833\text{ A}} = \boxed{144\ \Omega}$$

(c) The bulbs are in parallel, so

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{96.0\ \Omega} + \frac{1}{96.0\ \Omega} + \frac{1}{144\ \Omega}$$

$$R_{\text{eq}} = \boxed{36.0\ \Omega}$$

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Section 33.3 Inductors in an AC Circuit

P33.9 Inductive reactance is proportional to frequency.

At 50.0 Hz,

$$\frac{X'_L}{X_L} = \frac{f'}{f}$$

$$X'_L = X_L \frac{f'}{f} = \frac{50.0\,\text{Hz}}{60.0\,\text{Hz}}(54.0\ \Omega) = 45.0\ \Omega$$

The maximum current is

$$I_{\max} = \frac{\Delta V_{\max}}{X_L} = \frac{\sqrt{2}\left(\Delta V_{\text{rms}}\right)}{X_L} = \frac{\sqrt{2}\,(100\ \text{V})}{45.0\ \Omega} = \boxed{3.14\ \text{A}}$$

P33.10 (a) $X_L = \omega L = \dfrac{\Delta V_{\max}}{I_{\max}}$

$$L = \frac{\Delta V_{\max}}{\omega I_{\max}} = \frac{100\ \text{V}}{2\pi(50.0\ \text{Hz})(7.50\ \text{A})} = \boxed{0.0424\ \text{H}}$$

(b) From $I_{\max} = \dfrac{\Delta V_{\max}}{X_L} = \dfrac{\Delta V_{\max}}{\omega L}$, we see that is current inversely proportional to angular frequency:

$$\frac{I_{\max}}{I'_{\max}} = \frac{\omega'}{\omega}$$

$$\omega' = \omega \frac{I_{\max}}{I'_{\max}} = \left[2\pi(50.0\ \text{Hz})\right]\frac{7.50\ \text{A}}{2.50\ \text{A}} = \boxed{942\ \text{rad/s}}$$

P33.11 The inductive reactance is

$$X_L = \omega L = (65.0\ \pi\ \text{s}^{-1})(70.0 \times 10^{-3}\ \text{V}\cdot\text{s/A}) = 14.3\ \Omega$$

The amplitude of the current is

$$I_{\max} = \frac{\Delta V_{\max}}{X_L} = \frac{80.0\ \text{V}}{14.3\ \Omega} = 5.60\ \text{A}$$

Equation 33.7 lets us evaluate the current:

$$i = -I_{\max}\cos\omega t = -(5.60\ \text{A})\cos\left[\left(65.0\pi\ \text{s}^{-1}\right)(0.015\ 5\ \text{s})\right]$$

$$= -(5.60\ \text{A})\cos(3.17\ \text{rad}) = \boxed{+5.60\ \text{A}}$$

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P33.12 The relationship between current, inductance, and maximum voltage is

$$I_{\text{rms}} = \frac{\Delta V_{L,\text{rms}}}{X_L} = \frac{\left(\Delta V_{L,\max}/\sqrt{2}\right)}{\omega L} = \frac{\Delta V_{L,\max}}{\sqrt{2}(2\pi) fL}$$

In order to restrict the current to $I_{\text{rms}} < 2.00 \times 10^{-3}\ \text{A},$ we require

$$\frac{\Delta V_{L,\max}}{\sqrt{2}(2\pi) fL} < 2.00 \times 10^{-3}\ \text{A}$$

Solving for the inductance then gives

$$L > \frac{\Delta V_{L,\max}}{\sqrt{2}(2\pi) f\left(2.00 \times 10^{-3}\ \text{A}\right)}$$

$$= \frac{4.00\ \text{V}}{\sqrt{2}(2\pi)(300.0\ \text{Hz})\left(2.00 \times 10^{-3}\ \text{A}\right)}$$

or $\quad L > \boxed{0.750\ \text{H}}$

P33.13 (a) $\quad X_L = 2\pi f L = 2\pi(80.0\ \text{Hz})\left(25.0 \times 10^{-3}\ \text{H}\right) = \boxed{12.6\ \Omega}$

(b) $\quad I_{\text{rms}} = \dfrac{\Delta V_{L,\text{rms}}}{X_L} = \dfrac{78.0\ \text{V}}{X_L} = \boxed{6.21\ \text{A}}$

(c) $\quad I_{\max} = \sqrt{2}\, I_{\text{rms}} = \sqrt{2}(6.21\ \text{A}) = \boxed{8.78\ \text{A}}$

P33.14 In the inductor, because $U_B = \dfrac{1}{2} L i_L^2 = 0$ when $t = 0,\ i_L = I_{\max} \sin(\omega t).$ Then,

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{X_L} = \frac{\Delta V_{\text{rms}}}{\omega L} = \frac{120\ \text{V}}{\left[2\pi(60.0)\ \text{s}^{-1}\right](0.020\,0\ \text{H})} = 15.9\ \text{A}$$

and $\quad I_{\max} = \sqrt{2} I_{\text{rms}} = \sqrt{2}(15.9\ \text{A}) = 22.5\ \text{A}$

the current in the inductor is

$$i_L = I_{\max} \sin \omega t = (22.5\ \text{A}) \sin\left[2\pi(60.0)\ \text{s}^{-1} \cdot \left(\frac{1}{180}\ \text{s}\right)\right]$$

$$= (22.5\ \text{A}) \sin 120^\circ = 19.5\ \text{A}$$

and the energy stored is

$$U_B = \frac{1}{2} L i_L^2 = \frac{1}{2}(0.020\,0\ \text{H})(19.5\ \text{A})^2 = \boxed{3.80\ \text{J}}$$

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P33.15 The flux Φ_B through each turn of the inductor is related to the inductance by

$$L = \frac{N\Phi_B}{i}$$

Then, for an N-turn inductor,

$$N\Phi_{B,\max} = LI_{\max} = \frac{X_L}{\omega}\frac{\left(\Delta V_{L,\max}\right)}{X_L} = \frac{\left(\Delta V_{L,\max}\right)}{\omega}$$

$$N\Phi_{B,\max} = \frac{\sqrt{2}\left(\Delta V_{L,\text{rms}}\right)}{2\pi f} = \frac{120\text{ V}}{\sqrt{2}\pi\left(60.0\text{ s}^{-1}\right)} = \boxed{0.450\text{ Wb}}$$

P33.16 We are given: $\Delta v = 120\sin 30.0\pi t$ where Δv is in volts and t in seconds, and L = 0.500 H.

(a) By inspection, $\omega = 30\pi$ rad/s , so

$$f = \frac{\omega}{2\pi} = \frac{30\pi\text{ rad/s}}{2\pi} = \boxed{15.0\text{ Hz}}.$$

(b) Also by inspection, $\Delta V_{L,\max} = 120\text{ V}$, so that

$$\Delta V_{L,\text{rms}} = \frac{\Delta V_{L,\max}}{\sqrt{2}} = \frac{120\text{ V}}{\sqrt{2}} = \boxed{84.9\text{ V}}$$

(c) $X_L = 2\pi fL = \omega L = \left(30\pi\text{ rad/s}\right)\left(0.500\text{ H}\right) = \boxed{47.1\ \Omega}$

(d) $I_{\text{rms}} = \frac{\Delta V_{L,\text{rms}}}{X_L} = \frac{84.9\text{ V}}{47.1\ \Omega} = \boxed{1.80\text{ A}}$

(e) $I_{\max} = \sqrt{2}\,I_{\text{rms}} = \sqrt{2}\left(1.80\text{ A}\right) = \boxed{2.55\text{ A}}$

Section 33.4 Capacitors in an AC Circuit

P33.17 Current leads voltage by 90° in a capacitor, and because charge is proportional to voltage, current leads charge by 90°. If $\Delta v_C = \Delta V_{\max}\sin\omega t$, then $q = C\left(\Delta V_{\max}\right)\sin\omega t$ so that the stored energy is $U_C = \frac{q^2}{2C} = 0$ when t = 0. Therefore, the current is given by

$$i_C = I_{\max}\sin\left(\omega t + 90°\right) = \frac{\Delta V_{\max}}{X_C}\sin\left(\omega t + 90°\right)$$

The capacitive reactance is

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$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi(60.0/\text{s})(1.00\times10^{-3}\ \text{C/V})} = 2.65\ \Omega$$

and the current at $t = \frac{1}{180}\ \text{s}$ is

$$\begin{aligned} i_C &= \frac{\Delta V_{\max}}{X_C}\sin(\omega t + \phi) \\ &= \frac{\sqrt{2}(120\ \text{V})}{2.65\ \Omega}\sin\left[2\pi\left(60.0\ \text{s}^{-1}\right)\cdot\left(\frac{1}{180}\ \text{s}\right)+\frac{\pi}{2}\right] \\ &= -32.0\ \text{A} \end{aligned}$$

The magnitude of the current is $\boxed{32.0\ \text{A}}$.

P33.18 (a) $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(60.0\ \text{Hz})(12.0\times10^{-6}\ \text{F})} = \boxed{221\ \Omega}$

(b) $I_{\text{rms}} = \frac{\Delta V_{C,\text{rms}}}{X_C} = \frac{36.0\ \text{V}}{221\ \Omega} = \boxed{0.163\ \text{A}}$

(c) $I_{\max} = \sqrt{2}\, I_{\text{rms}} = \boxed{0.230\ \text{A}}$

(d) $\boxed{\text{No.}}$ Current leads voltage, and thus charge, by 90° in a capacitor. The current reaches its maximum value one-quarter cycle before the voltage reaches its maximum value. From the definition of capacitance, the capacitor reaches its maximum charge when the voltage across it is also a maximum. Consequently, the maximum charge and the maximum current do not occur at the same time.

P33.19 (a) We require $X_C = \frac{1}{2\pi fC} < 175\ \Omega$, or

$$\frac{1}{2\pi f\left(22.0\times10^{-6}\ \text{F}\right)} < 175\ \Omega$$

Solving,

$$\frac{1}{2\pi(22.0\times10^{-6}\ \text{F})(175\ \Omega)} < f$$

or $\boxed{f > 41.3\ \text{Hz}}$

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(b) As a function of capacitance C, $X_C(C) \propto \dfrac{1}{C}$, so

$$X_C(\text{C} = 44.0\ \mu\text{F}) = \frac{1}{2} X_C(\text{C} = 22.0\ \mu\text{F})$$

or new $X_C = \dfrac{1}{2}$ old X_C.

Therefore, $\boxed{X_C < 87.5\ \Omega}$

P33.20 (a) By inspection, $\Delta V_{C,\max} = 98.0$ V, so

$$\Delta V_{C,\text{rms}} = \frac{\Delta V_{C,\max}}{\sqrt{2}} = \frac{98.0\ \text{V}}{\sqrt{2}} = \boxed{69.3\ \text{V}}.$$

(b) Also by inspection, $\omega = 80\pi$ rad/s , so

$$f = \frac{\omega}{2\pi} = \frac{80\pi\ \text{rad/s}}{2\pi\ \text{rad}} = \boxed{40.0\ \text{Hz}}$$

(c) We can find the capacitive reactance from

$$X_C = \frac{\Delta V_{C,\max}}{I_{\max}} = \frac{98.0\ \text{V}}{0.500\ \text{A}} = 196\ \Omega$$

and since

$$X_C = \frac{1}{2\pi f C} = \frac{1}{\omega C}$$

solving for the capacitance gives

$$C = \frac{1}{\omega X_C} = \frac{1}{(80\pi\ \text{rad/s})(196\ \Omega)} = 2.03 \times 10^{-5}\ \text{F} = \boxed{20.3\ \mu\text{F}}$$

P33.21 We combine the steps in the equation

$$I_{\max} = \frac{\Delta V_{\max}}{X_C} = \Delta V_{\max} \omega C = \Delta V_{\max} (2\pi f C)$$

Then,

$$I_{\max} = (48.0\ \text{V})(2\pi)(90.0\ \text{Hz})\left(3.70 \times 10^{-6}\ \text{F}\right) = \boxed{100\ \text{mA}}$$

P33.22 The maximum charge is given by

$$Q_{\max} = C(\Delta V_{\max}) = C\left[\sqrt{2}\,(\Delta V_{\text{rms}})\right] = \boxed{\sqrt{2} C (\Delta V_{\text{rms}})}$$

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P33.23 The maximum current in the capacitor is given by

$$I_{\max} = \sqrt{2} I_{\text{rms}} = \frac{\sqrt{2}\left(\Delta V_{\text{rms}}\right)}{X_C} = \sqrt{2}\left(\Delta V_{\text{rms}}\right) 2\pi f C$$

(a) For the North American electrical outlet,

$$I_{\max} = \sqrt{2}\,(120\text{ V})\,2\pi\left(60.0/\text{s}\right)\left(2.20\times10^{-6}\text{ C/V}\right) = \boxed{141\text{ mA}}$$

(b) For the European electrical outlet,

$$I_{\max} = \sqrt{2}\,(240\text{ V})\,2\pi\left(50.0/\text{s}\right)\left(2.20\times10^{-6}\text{ F}\right) = \boxed{235\text{ mA}}$$

Section 33.5 The *RLC* Series Circuit

P33.24 We first determine the reactances of the circuit. The capacitive reactance is

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi(50.0)\left(65.0\times10^{-6}\text{ F}\right)} = 49.0\ \Omega$$

the inductive reactance is,

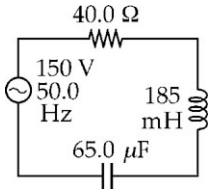
$$X_L = \omega L = 2\pi(50.0)\left(185\times10^{-3}\text{ H}\right) = 58.1\ \Omega$$

and the impedance Z of the circuit is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{(40.0\ \Omega)^2 + (58.1\ \Omega - 49.0\ \Omega)^2}$$
$$= 41.0\ \Omega$$

The current in the circuit is then

$$I_{\max} = \frac{\Delta V_{\max}}{Z} = \frac{150\text{ V}}{41.0\ \Omega} = 3.66\text{ A}$$



ANS. FIG. P33.24

(a) The maximum voltage between points *a* and *b* is the potential drop across the resistor:

$$\Delta V_R = I_{\max} R = (3.66\text{ A})(40.0\ \Omega) = \boxed{146\text{ V}}$$

(b) The maximum voltage between points *b* and *c* is the potential

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drop across the coil:

$$\Delta V_L = I_{\max} X_L = (3.66\ \text{A})(58.1\ \Omega) = 212.5\ \text{V} = \boxed{212\ \text{V}}$$

(c) The maximum voltage between points *c* and *d* is the potential drop across the capacitor:

$$\Delta V_C = I_{\max} X_C = (3.66\ \text{A})(49.0\ \Omega) = 179.1\ \text{V} = \boxed{179\ \text{V}}$$

(d) The potential drop between points *b* and *d* is

$$\Delta V_L - \Delta V_C = 212.5\ \text{V} - 179.1\ \text{V} = \boxed{33.4\ \text{V}}$$

P33.25 The resistance of the circuit is $R = 300\ \Omega$. The inductive reactance of the circuit is

$$X_L = \omega L = 2\pi\left(\frac{500}{\pi}\ \text{s}^{-1}\right)(0.200\ \text{H}) = 200\ \Omega$$

The capacitive reactance of the circuit is

$$X_C = \frac{1}{\omega C} = \left[2\pi\left(\frac{500}{\pi}\ \text{s}^{-1}\right)\left(11.0\times10^{-6}\ \text{F}\right)\right]^{-1} = 90.9\ \Omega$$

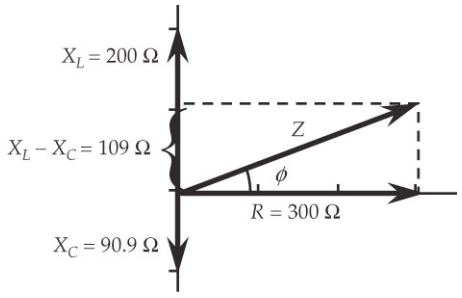
The impedance of the circuit is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{(300\ \Omega)^2 + (200\ \Omega - 90.0\ \Omega)^2} = 319\ \Omega$$

and

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{200\ \Omega - 90.9\ \Omega}{300\ \Omega}\right) = 20.0°$$

The phasor diagram is shown in ANS. FIG. P33.25.



ANS. FIG. P33.25

P33.26 (a) From the time dependence of the voltage, we recognize that $\omega = 100\ \text{s}^{-1}$. The resistance of the circuit is $R = 68.0\ \Omega$, the inductive reactance of the circuit is

$$X_L = \omega L = \left(100\ \text{s}^{-1}\right)(0.160\ \text{H}) = 16.0\ \Omega$$

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The capacitive reactance of the circuit is

$$X_C = \frac{1}{\omega C} = \frac{1}{\left(100\ \text{s}^{-1}\right)\left(99.0\times10^{-6}\ \text{F}\right)} = 101\ \Omega$$

Therefore, the impedance of the circuit is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}$$

$$= \sqrt{(68.0\ \Omega)^2 + (16.0\ \Omega - 101\ \Omega)^2} = \boxed{109\ \Omega}$$

(b) The maximum current in the circuit is

$$I_{\max} = \frac{\Delta V_{\max}}{Z} = \frac{40.0\ \text{V}}{109\ \Omega} = \boxed{0.367\ \text{A}}$$

(c) $\boxed{\omega = 100\ \text{rad/s}}$

(d) We find ϕ from

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{16.0\ \Omega - 101\ \Omega}{68.0\ \Omega}\right)$$

$$= -0.896\ \text{rad} = \boxed{-51.3°}$$

P33.27 (a) The inductive reactance of the circuit is

$$X_L = \omega L = 2\pi\left(50.0\ \text{s}^{-1}\right)\left(150\times10^{-3}\ \text{H}\right) = \boxed{47.1\ \Omega}$$

(b) The capacitive reactance of the circuit is

$$X_C = \frac{1}{\omega C} = \left[2\pi\left(50.0\ \text{s}^{-1}\right)\left(5.00\times10^{-6}\ \text{F}\right)\right]^{-1} = \boxed{637\ \Omega}$$

(c) The impedance of the circuit is

$$Z = \frac{\Delta V_{\max}}{I_{\max}} = \frac{240\ \text{V}}{0.100\ \text{A}} = 2.40\times10^3\ \Omega = \boxed{2.40\ \text{k}\Omega}$$

(d) From the definition of impedance,

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}$$

Solving for the resistance gives

$$R = \sqrt{Z^2 - \left(X_L - X_C\right)^2}$$

$$= \sqrt{\left(2.40\times10^3\ \Omega\right)^2 - (47.1\ \Omega - 637\ \Omega)^2}$$

$$= 2.33\times10^3\ \Omega = \boxed{2.33\ \text{k}\Omega}$$

(e) $\phi = \tan^{-1}\left[\frac{X_L - X_C}{R}\right] = \tan^{-1}\left(\frac{47.1\ \Omega - 637\ \Omega}{2.33\times10^3\ \Omega}\right) = \boxed{-14.2°}$

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P33.28 From the definitions of inductive and capacitive reactance, $X_L = \omega L$ and $X_C = \dfrac{1}{\omega C}$. Setting these equal and solving for the angular frequency gives

$$\omega L = \frac{1}{\omega C}$$

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\left(57.0\times10^{-6}\right)\left(57.0\times10^{-6}\right)}} = 1.75\times10^{4} \text{ rad/s}$$

Then,

$$f = \frac{\omega}{2\pi} = \boxed{2.79 \text{ kHz}}$$

P33.29 The reactance of the inductor is

$$X_L = 2\pi f L = 2\pi\left(60.0 \text{ s}^{-1}\right)(0.460 \text{ H}) = 173 \ \Omega$$

The reactance of the capacitor is

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} = \frac{1}{2\pi\left(60.0 \text{ s}^{-1}\right)\left(21.0 \times 10^{-6} \text{ F}\right)} = 126 \ \Omega$$

(a) $\phi = \tan^{-1}\left(\dfrac{X_L - X_C}{R}\right) = \tan^{-1}\left(\dfrac{173 \ \Omega - 126 \ \Omega}{150 \ \Omega}\right) = \boxed{17.4°}$

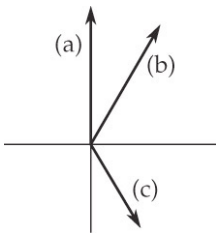
(b) Since $X_L > X_C$, ϕ is positive, so $\boxed{\text{voltage leads the current}}$. This means that the power-supply or total voltage goes through each maximum, zero-crossing, and minimum earlier in time than the current does.

P33.30 The Phasors for the three cases are shown in ANS. FIG. P33.30.

(a) 25.0 sin ωt at $\omega t = 90.0°$

(b) 30.0 sin ωt at $\omega t = 60.0°$

(c) 18.0 sin ωt at $\omega t = 300°$



ANS. FIG. P33.30

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P33.31 (a) We first find the impedances of the inductor and the capacitor:

$$X_L = \omega L = 2\pi(50.0)\left(400\times10^{-3}\right) = 126\ \Omega$$

and $$X_C = \frac{1}{\omega C} = \frac{1}{2\pi(50.0)\left(4.43\times10^{-6}\right)} = 719\ \Omega$$

We then compute the impedance of the circuit:

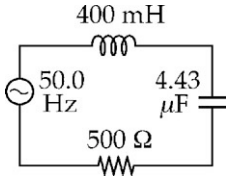
$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{500^2 + (126-719)^2} = 776\ \Omega$$

Then, from the equation for a series RLC circuit,

$$\Delta V_{\max} = I_{\max} Z = \left(250\times10^{-3}\right)(776) = \boxed{194\ \text{V}}$$

(b) $$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{126-719}{500}\right) = \boxed{-49.9°}$$

Thus, the $\boxed{\text{current leads the voltage}}$.



ANS. FIG. P33.31

P33.32 (a) The capacitive reactance of the circuit is

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(60.0\ \text{Hz})\left(30.0\times10^{-6}\ \text{F}\right)} = \boxed{88.4\ \Omega}$$

(b) The impedance of the circuit is

$$Z = \sqrt{R^2 + \left(0 - X_C\right)^2} = \sqrt{R^2 + X_C^2} = \sqrt{(60.0\ \Omega)^2 + (88.4\ \Omega)^2}$$

$$= \boxed{107\ \Omega}$$

(c) $$I_{\max} = \frac{\Delta V_{\max}}{Z} = \frac{1.20\times10^2\ \text{V}}{107\ \Omega} = \boxed{1.12\ \text{A}}$$

(d) The phase angle in this *RC* circuit is

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{0 - 88.4\ \Omega}{60.0\ \Omega}\right) = -55.8°$$

Since $\phi < 0$, $\boxed{\text{the voltage lags behind the current by } 55.8°}$.

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(e) Adding an inductor will change the impedance, and hence the current in the circuit. The current could be larger or smaller, depending on the inductance added. The largest current would result when the inductive reactance equals the capacitive reactance, the impedance has its minimum value, equal to 60.0 Ω, and the current in the circuit is

$$I_{\max} = \frac{\Delta V_{\max}}{Z} = \frac{\Delta V_{\max}}{R} = \frac{1.20 \times 10^2 \text{ V}}{60.0\ \Omega} = 2.00 \text{ A}$$

P33.33 Let X_C represent the initial capacitive reactance. Moving the plates to half their original separation doubles the capacitance ($C = \frac{\epsilon_0 A}{d}$) and cuts $X_C = \frac{1}{\omega C}$ in half.

For the current to double, the total impedance must be cut in half, or $Z_i = 2Z_f$. Then,

$$\sqrt{R^2 + \left(X_L - X_C\right)^2} = 2\sqrt{R^2 + \left(X_L - \frac{X_C}{2}\right)^2}$$

$$R^2 + \left(R - X_C\right)^2 = 4\left[R^2 + \left(R - \frac{X_C}{2}\right)^2\right]$$

$$2R^2 - 2RX_C + X_C^2 = 8R^2 - 4RX_C + X_C^2$$

$$\boxed{X_C = 3R}$$

Section 33.6 Power in an AC Circuit

P33.34 The power factor for a series *RLC* circuit is given by

$$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + \left(X_L - X_C\right)^2}}$$

The circuit in this problem has no capacitance, so the power factor becomes

$$\cos\phi = \frac{R}{\sqrt{R^2 + X_L{}^2}}$$

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In order for the power factor to be equal to 1.00, we would have to have $X_L = 0$, which would require either L or f to be zero. Because this is not the case, the situation is impossible.

P33.35 From the definition of impedance, $Z = \sqrt{R^2 + (X_L - X_C)^2}$, we have

$$(X_L - X_C) = \sqrt{Z^2 - R^2}$$

Substituting numerical values,

$$(X_L - X_C) = \sqrt{(75.0\ \Omega)^2 - (45.0\ \Omega)^2} = 60.0\ \Omega$$

The phase angle of the circuit is then

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{60.0\ \Omega}{45.0\ \Omega}\right) = 53.1°$$

The rms current in the circuit is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{210\ \text{V}}{75.0\ \Omega} = 2.80\ \text{A}$$

Therefore, the power delivered to the circuit is

$$P = (\Delta V_{\text{rms}}) I_{\text{rms}} \cos\phi = (210\ \text{V})(2.80\ \text{A})\cos(53.1°) = \boxed{353\ \text{W}}$$

P33.36 The rms voltage of the power supply is

$$\Delta V_{\text{rms}} = \frac{100\ \text{V}}{\sqrt{2}} = 70.7\ \text{V}$$

In order to calculate the impedance, we first need the capacitive and inductive reactances:

$$X_C = \frac{1}{\omega C} = \frac{1}{(1\ 000\ \text{s}^{-1})(5.00 \times 10^{-6}\ \text{F})} = 200\ \Omega$$

$$X_L = \omega L = (1\ 000\ \text{s}^{-1})(0.500\ \text{H}) = 500\ \Omega$$

Next,

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$
$$= \sqrt{(400\ \Omega)^2 + (500\ \Omega - 200\ \Omega)^2} = 500\ \Omega$$

The rms current is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{70.7\ \text{V}}{500\ \Omega} = 0.141\ \text{A}$$

The average power is $P_{\text{avg}} = I_{\text{rms}}^2 R = (0.141\ \text{A})^2(400\ \Omega) = \boxed{8.00\ \text{W}}$

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P33.37 The rms current in the circuit is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{160\ \text{V}}{80.0\ \Omega} = 2.00\ \text{A}$$

and the average power delivered to the circuit is

$$P_{\text{avg}} = I_{\text{rms}}^2 R = (2.00\ \text{A})^2 (22.0\ \Omega) = \boxed{88.0\ \text{W}}$$

P33.38 Given $v = \Delta V_{\max} \sin(\omega t) = (90.0\ \text{V})\sin(350t)$, observe that $\Delta V_{\max} = 90.0\ \text{V}$ and $\omega = 350\ \text{rad/s}$. Also, the net reactance is

$$X_L - X_C = 2\pi f L - \frac{1}{2\pi f C} = \omega L - \frac{1}{\omega C}$$

(a) To find the impedance, we first compute

$$\begin{aligned} X_L - X_C &= \omega L - \frac{1}{\omega C} \\ &= (350\ \text{rad/s})(0.200\ \text{H}) - \frac{1}{(350\ \text{rad/s})(25.0\times 10^{-6}\ \text{F})} \\ &= -44.3\ \Omega \end{aligned}$$

so the impedance is

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(50.0\ \Omega)^2 + (-44.3\ \Omega)^2} = \boxed{66.8\ \Omega}$$

(b) The rms current in the circuit is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{\Delta V_{\max}/\sqrt{2}}{Z} = \frac{90.0\ \text{V}}{\sqrt{2}(66.8\ \Omega)} = \boxed{0.953\ \text{A}}$$

(c) The phase difference between the applied voltage and the current is

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \tan^{-1}\left(\frac{-44.3\ \Omega}{50.0\ \Omega}\right) = -41.5°$$

so the average power delivered to the circuit is

$$\begin{aligned} P_{\text{avg}} &= I_{\text{rms}} \Delta V_{\text{rms}} \cos\phi = I_{\text{rms}} \left(\frac{\Delta V_{\max}}{\sqrt{2}}\right)\cos\phi \\ &= (0.953\ \text{A})\left(\frac{90.0\ \text{V}}{\sqrt{2}}\right)\cos(-41.5°) = \boxed{45.4\ \text{W}} \end{aligned}$$

P33.39 The power is given by

$$P_{\text{avg}} = I_{\text{rms}} \Delta V_{\text{rms}} \cos\phi = I_{\text{rms}}^2 R$$

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(a) Then,

$$P = I_{\text{rms}}\left(\Delta V_{\text{rms}}\right)\cos\phi = (9.00\text{ A})(180\text{ V})\cos(-37.0^\circ)$$
$$= 1.29\times 10^3\text{ W}$$

Then, from $P = I_{\text{rms}}^2 R$,

$$R = \frac{P}{I_{\text{rms}}^2} = \frac{1.29\times 10^3\text{ W}}{(9.00\text{ A})^2} = \boxed{16.0\ \Omega}$$

(b) From the definition of phase angle, $\tan\phi = \dfrac{X_L - X_C}{R}$,

$$X_L - X_C = R\tan\phi = (16.0\ \Omega)\tan(-37.0^\circ) = \boxed{-12.0\ \Omega}$$

P33.40 For this circuit, R = 20.0 Ω, the capacitive reactance is $X_C = 0$, and the inductive reactance is

$$X_L = \omega L = 2\pi\left(60.0\text{ s}^{-1}\right)\left(0.025\,0\text{ H}\right) = 9.42\ \Omega$$

The impedance of the circuit is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{(20.0\ \Omega)^2 + (9.42\ \Omega)^2} = 22.1\ \Omega$$

(a) The rms current in the circuit is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{120\text{ V}}{22.1\ \Omega} = \boxed{5.43\text{ A}}$$

(b) The phase angle of the circuit is

$$\phi = \tan^{-1}\left(\frac{9.42}{20.0}\right) = 25.2^\circ$$

so the power factor is

$$\cos\phi = \boxed{0.905}$$

(c) For the power factor to equal 1, we require $\phi = 0$, and this can only occur if $X_L = X_C$, or

$$9.42\ \Omega = \frac{1}{2\pi\left(60.0\text{ s}^{-1}\right)C} \rightarrow C = \boxed{281\ \mu\text{F}}$$

(d) For the power to equal that before the capacitor was installed, or $P_b = P_d$, we require

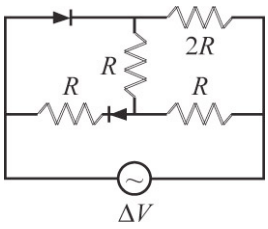
$$\left(\Delta V_{\text{rms}}\right)_b \left(I_{\text{rms}}\right)_b \cos\phi_b = \frac{\left(\Delta V_{\text{rms}}\right)_d^2}{R}$$

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Solving for the rms voltage gives

$$\left(\Delta V_{\text{rms}}\right)_d = \sqrt{R\left(\Delta V_{\text{rms}}\right)_b \left(I_{\text{rms}}\right)_b \cos\phi_b}$$
$$= \sqrt{(20.0\ \Omega)(120\ \text{V})(5.43\ \text{A})(0.905)}$$
$$= \boxed{109\ \text{V}}$$

P33.41 One-half of the time, the left side of the generator is positive, the top diode conducts, and the bottom diode switches off. The power supply sees resistance



ANS. FIG. P33.41

$$\left[\frac{1}{2R} + \frac{1}{2R}\right]^{-1} = R$$

and the power is $\dfrac{\left(\Delta V_{\text{rms}}\right)^2}{R}$.

The other half of the time the right side of the generator is positive, the upper diode is an open circuit, and the lower diode has zero resistance. The equivalent resistance is then

$$R_{\text{eq}} = R + \left[\frac{1}{3R} + \frac{1}{R}\right]^{-1} = \frac{7R}{4}$$

and
$$P = \frac{\left(\Delta V_{\text{rms}}\right)^2}{R_{\text{eq}}} = \frac{4\left(\Delta V_{\text{rms}}\right)^2}{7R}$$

The overall time average power is:

$$P_{\text{avg}} = \frac{\left[\left(\Delta V_{\text{rms}}\right)^2 / R\right] + \left[4\left(\Delta V_{\text{rms}}\right)^2 / 7R\right]}{2} = \boxed{\frac{11\left(\Delta V_{\text{rms}}\right)^2}{14R}}$$

Section 33.7 Resonance in a Series *RLC* Circuit

P33.42 We are given $L = 0.020\ 0$ H, $C = 100 \times 10^{-9}$ F, $R = 20.0\ \Omega$, and $\Delta V_{\text{max}} = 100$ V.

(a) The resonant frequency for a series *RLC* circuit is

$$f = \frac{\omega_0}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{1}{LC}} = \boxed{3.56\ \text{kHz}}$$

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(b) At resonance,

$$I_{\max} = \frac{\Delta V_{\max}}{R} = \boxed{5.00\ \text{A}}$$

(c) From Equation 33.38,

$$Q = \frac{\omega_0 L}{R} = \boxed{22.4}$$

(d) At resonance, the amplitude of the voltage across the inductor is

$$\Delta V_{L,\,\max} = X_L I_{\max} = \omega_0 L I_{\max} = \boxed{2.24\ \text{kV}}$$

P33.43 The circuit is to be in resonance when

$$\omega_0 L = \frac{1}{\omega_0 C}$$

Solving for the capacitance gives

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4\pi^2 (99.7\ \text{MHz})^2 (1.40\ \mu\text{V}\cdot\text{s/A})}$$
$$= \boxed{1.82\ \text{pF}}$$

P33.44 (a) The resonance frequency of a *RLC* circuit is $f_0 = 1/2\pi\sqrt{LC}$. Thus, the inductance is

$$L = \frac{1}{4\pi^2 f_0^2 C} = \frac{1}{4\pi^2 \left(9.00\times10^{9}\ \text{Hz}\right)^2 \left(2.00\times10^{-12}\ \text{F}\right)}$$
$$= 1.56\times10^{-10}\ \text{H} = \boxed{156\ \text{pH}}$$

(b) At resonance,

$$X_L = X_C = \frac{1}{2\pi f_0 C} = \frac{1}{2\pi \left(9.00\times10^{9}\ \text{Hz}\right)\left(2.00\times10^{-12}\ \text{F}\right)}$$
$$= \boxed{8.84\ \Omega}$$

P33.45 The resonance frequency is $\omega_0 = \dfrac{1}{\sqrt{LC}}$. Thus, if $\omega = 2\omega_0$, then

$$X_L = \omega L = \left(\frac{2}{\sqrt{LC}}\right) L = 2\sqrt{\frac{L}{C}}$$

and $$X_C = \frac{1}{\omega C} = \frac{\sqrt{LC}}{2C} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

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The impedance of the circuit is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{R^2 + 2.25\left(\frac{L}{C}\right)}$$

so the rms current in the circuit is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{\Delta V_{\text{rms}}}{\sqrt{R^2 + 2.25\left(L/C\right)}}$$

The power delivered to the circuit is

$$P = \left(I_{\text{rms}}\right)^2 R = \left(\frac{\Delta V_{\text{rms}}}{Z}\right)^2 R = \frac{\left(\Delta V_{\text{rms}}\right)^2}{Z^2} R = \frac{\left(\Delta V_{\text{rms}}\right)^2 R}{R^2 + 2.25\left(L/C\right)}$$

and the energy delivered in one period is $E = P\Delta t$:

$$E = P\Delta t = \frac{\left(\Delta V_{\text{rms}}\right)^2 R}{R^2 + 2.25\left(L/C\right)}\left(\frac{2\pi}{\omega}\right) = \frac{\left(\Delta V_{\text{rms}}\right)^2 RC}{R^2 C + 2.25L}\left(\pi\sqrt{LC}\right)$$

$$= \frac{4\pi\left(\Delta V_{\text{rms}}\right)^2 RC\sqrt{LC}}{4R^2 C + 9L}$$

Substituting numerical values,

$$E = \frac{4\pi\left(50.0\,\text{V}\right)^2 (10.0\ \Omega)\left(100\times10^{-6}\ \text{F}\right)\left[\left(10.0\times10^{-3}\ \text{H}\right)\left(100\times10^{-6}\ \text{F}\right)\right]^{1/2}}{4(10.0\ \Omega)^2\left(100\times10^{-6}\ \text{F}\right) + 9\left(10.0\times10^{-3}\ \text{H}\right)}$$

$$= \boxed{242\ \text{mJ}}$$

P33.46 The resonance frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$. Thus, if $\omega = 2\omega_0$, then

$$X_L = \omega L = \left(\frac{2}{\sqrt{LC}}\right)L = 2\sqrt{\frac{L}{C}}$$

and

$$X_C = \frac{1}{\omega C} = \frac{\sqrt{LC}}{2C} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

The impedance of the circuit is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{R^2 + 2.25\left(\frac{L}{C}\right)}$$

so the rms current in the circuit is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{\Delta V_{\text{rms}}}{\sqrt{R^2 + 2.25\left(L/C\right)}}$$

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The power delivered to the circuit is

$$P = \left(I_{\text{rms}}\right)^2 R = \left(\frac{\Delta V_{\text{rms}}}{Z}\right)^2 R = \frac{\left(\Delta V_{\text{rms}}\right)^2}{Z^2} R = \frac{\left(\Delta V_{\text{rms}}\right)^2 R}{R^2 + 2.25\left(L/C\right)}$$

and the energy delivered in one period is $E = P\Delta t$:

$$E = P\Delta t = \frac{\left(\Delta V_{\text{rms}}\right)^2 R}{R^2 + 2.25\left(L/C\right)}\left(\frac{2\pi}{\omega}\right) = \frac{\left(\Delta V_{\text{rms}}\right)^2 RC}{R^2 C + 2.25L}\left(\pi\sqrt{LC}\right)$$

$$= \boxed{\frac{4\pi\left(\Delta V_{\text{rms}}\right)^2 RC\sqrt{LC}}{4R^2 C + 9L}}$$

P33.47 (a) To find the capacitance, we note that $f = \dfrac{1}{2\pi\sqrt{LC}}$. Solving for the capacitance C gives

$$C = \frac{1}{4\pi^2 f^2 L}$$

$$= \frac{1}{4\pi^2\left(1.00\times10^{10}\ \text{Hz}\right)^2\left(400\times10^{-12}\ \text{H}\right)}$$

$$= 6.33\times10^{-13}\ \text{F} = \boxed{0.633\ \text{pF}}$$

(b) From Equation 26.15 for the capacitance of parallel plates with a dielectric, we have

$$C = \frac{\kappa\epsilon_0 A}{d} = \frac{\kappa\epsilon_0 \ell^2}{d}$$

Solving for the edge length gives

$$\ell = \left(\frac{Cd}{\kappa\epsilon_0}\right)^{1/2} = \left[\frac{\left(6.33\times10^{-13}\ \text{F}\right)\left(1.00\times10^{-3}\ \text{m}\right)}{(1)\left(8.85\times10^{-12}\ \text{F}\right)}\right]^{1/2}$$

$$= 8.46\times10^{-3}\ \text{m} = \boxed{8.46\ \text{mm}}$$

(c) The inductive reactance of the circuit at resonance, equal to the capacitive reactance, is

$$X_L = 2\pi f L = 2\pi\left(1.00\times10^{10}\ \text{Hz}\right)\left(400\times10^{-12}\ \text{H}\right) = \boxed{25.1\ \Omega}$$

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Section 33.8 The Transformer and Power Transmission

P33.48 (a) The output voltage is found from Equation 33.41, $\Delta v_2 = \frac{N_2}{N_1}\Delta v_1$. Therefore,

$$\Delta v_2 = \frac{1}{13}(120\text{ V}) = \boxed{9.23\text{ V}}$$

(b) Assuming an ideal transformer, $P_2 = P_1$. Therefore,

$$\Delta V_2 I_2 = \Delta V_1 I_1 = (120\text{ V})(0.0200\,\text{A}) = \boxed{2.40\text{ W}}$$

P33.49 The rms primary voltage is

$$\Delta V_{1,\text{rms}} = \frac{170\text{ V}}{\sqrt{2}} = 120\text{ V}$$

The rms voltage across the bigger coil is

$$\Delta V_{2,\text{rms}} = \left(\frac{N_2}{N_1}\right)\Delta V_{1,\text{rms}} = \left(\frac{2\,000}{350}\right)(120\text{ V}) = \boxed{687\text{ V}}$$

P33.50 (a) The total resistance of the transmission line is

$$R = \left(4.50\times10^{-4}\ \Omega/\text{m}\right)\left(6.44\times10^{5}\text{ m}\right) = 290\ \Omega$$

and the rms current in the line is

$$I_{\text{rms}} = \frac{P}{\Delta V_{\text{rms}}} = \frac{5.00\times10^{6}\text{ W}}{5.00\times10^{5}\text{ V}} = 10.0\text{ A}$$

The power loss during transmission is

$$P_{\text{loss}} = I_{\text{rms}}^2 R = (10.0\text{ A})^2(290\ \Omega) = \boxed{29.0\text{ kW}}$$

(b) The fraction of input power lost is

$$\frac{P_{\text{loss}}}{P} = \frac{2.90\times10^{4}\text{ W}}{5.00\times10^{6}\text{ W}} = \boxed{5.80\times10^{-3}}$$

(c) It is impossible to transmit so much power at such low voltage. Maximum power transfer occurs when load resistance equals the line resistance of 290 Ω, and is $\frac{\left(4.50\times10^{3}\text{ V}\right)^2}{2\cdot2(290\ \Omega)} = 17.5\text{ kW},$ far below the required 5 000 kW.

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P33.51 We find the voltage in the primary coil:

$$\frac{\Delta V_1}{\Delta V_2} = \frac{N_1}{N_2} \rightarrow \Delta V_1 = \Delta V_2 \frac{N_1}{N_2} = (25.0\text{ V})(2.50) = 62.5\text{ V}$$

The (average) power delivered to the load resistance is

$$P = I_2 \Delta V_2 = \frac{(\Delta V_2)^2}{R_2} = \frac{(25.0\text{ V})^2}{50.0\ \Omega} = 12.5\text{ W}$$

which is equal to the power delivered to the primary coil; thus, the (rms) current on the primary side is

$$I_1 = \frac{P}{\Delta V_1} = \frac{12.5\text{ W}}{62.5\text{ V}} = 0.200\text{ A}$$

On the primary side of the transformer, the voltages across the resistor and transformer (inductor) are 90° out of phase. Therefore,

$$(\Delta V_{\text{rms}})^2 = (\Delta V_{L,\text{rms}})^2 + (\Delta V_{R,\text{rms}})^2$$
$$(80.0\text{ V})^2 = (62.5\text{ V})^2 + (\Delta V_{R,\text{rms}})^2$$
$$\Delta V_{R,\text{rms}} = 49.9\text{ V}$$

and

$$\Delta V_{R,\text{rms}} = 49.9\text{ V} = I_{\text{rms}} R$$
$$R = \frac{\Delta V_{R,\text{rms}}}{I_{\text{rms}}} = \frac{49.9\text{ V}}{0.200\text{ A}} = \boxed{250\ \Omega}$$

P33.52 The capacitive reactance of this "circuit" is

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(60.0\text{ Hz})(20.0\times10^{-12}\text{ F})} = 1.33\times10^8\ \Omega$$

and the impedance is

$$Z = \sqrt{(50.0\times10^3\ \Omega)^2 + (1.33\times10^8\ \Omega)^2} \approx 1.33\times10^8\ \Omega$$

The rms current is then

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{5\,000\text{ V}}{1.33\times10^8\ \Omega} = 3.77\times10^{-5}\text{ A}$$

and the rms voltage across the person's body is

$$(\Delta V_{\text{rms}})_{\text{body}} = I_{\text{rms}} R_{\text{body}} = (3.77\times10^{-5}\text{ A})(50.0\times10^3\ \Omega)$$
$$= \boxed{1.88\text{ V}}$$

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Section 33.9 Rectifiers and Filters

P33.53 For this *RC* high-pass filter, the voltage gain ratio is

$$\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \frac{I_{\max} R}{I_{\max} Z} = \frac{R}{\sqrt{R^2 + X_C^2}}$$

With a capacitance of 613 μF and a frequency of 600 Hz, the capacitive reactance is

$$X_C = \frac{1}{2\pi(600\text{ Hz})\left(6.13\times10^{-4}\text{ F}\right)} = 0.433\ \Omega$$

and

$$\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \frac{R}{\sqrt{R^2 + X_C^2}} = \frac{0.500\ \Omega}{\sqrt{(0.500\ \Omega)^2 + (0.433\ \Omega)^2}} = \boxed{0.756}$$

P33.54 (a) The amplitude of the input voltage is

$$\Delta V_{\text{in}} = I_{\max} Z = I_{\max}\sqrt{R^2 + X_C^2} = I_{\max}\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

The amplitude of the output voltage is $\Delta V_{\text{out}} = I_{\max} R$. The gain ratio is

$$\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \frac{I_{\max} R}{I_{\max}\sqrt{R^2 + (1/\omega C)^2}} = \boxed{\frac{R}{\sqrt{R^2 + (1/\omega C)^2}}}$$

(b) As $\omega \to 0$, $\dfrac{1}{\omega C} \to \infty$, and $\dfrac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} \to \boxed{0}$.

(c) As $\omega \to \infty$, $\dfrac{1}{\omega C} \to 0$, and $\dfrac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} \to \dfrac{R}{R} = \boxed{1}$.

P33.55 (a) The input power is 8 W, and the useful output power is given by

$$I\Delta V = (0.3\text{ A})(9\text{ V}) = 2.7\text{ W}$$

The efficiency is then

$$\text{efficiency} = \frac{\text{useful output}}{\text{total input}} = \frac{2.7\text{ W}}{8\text{ W}} = 0.34 \to \boxed{34\%}$$

(b) Total input power = Total output power

8 W = 2.7 W + wasted power

wasted power = $\boxed{5.3\text{ W}}$

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(c) $E = P\Delta t = \left[6(8\text{ W})\right]\left[(31\text{ d})\left(\frac{86\,400\text{ s}}{1\text{ d}}\right)\right] = 1.29\times10^{8}\text{ J}\left(\frac{\$0.110}{3.6\times10^{6}\text{ J}}\right)$

$= \boxed{\$3.9}$

P33.56 (a) The amplitude of the input voltage is

$$\Delta V_{\text{in}} = IZ = I_{\max}\sqrt{R^2 + X_C^2} = I_{\max}\sqrt{R^2 + \left(1/\omega C\right)^2}$$

The amplitude of the output voltage is

$$\Delta V_{\text{out}} = I_{\max}X_C = \frac{I_{\max}}{\omega C}.$$

The gain ratio is therefore

$$\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} = \frac{I_{\max}/\omega C}{I_{\max}\sqrt{R^2 + \left(1/\omega C\right)^2}} = \boxed{\frac{1/\omega C}{\sqrt{R^2 + \left(1/\omega C\right)^2}}}$$

(b) As $\omega \to 0$, $\frac{1}{\omega C} \to \infty$ and R becomes negligible in comparison.

Then,

$$\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} \to \frac{1/\omega C}{1/\omega C} = \boxed{1}$$

(c) As $\omega \to \infty$, $\frac{1}{\omega C} \to 0$ and $\frac{\Delta V_{\text{out}}}{\Delta V_{\text{in}}} \to \boxed{0}$.

(d) We start with

$$\frac{1}{2} = \frac{1/\omega C}{\sqrt{R^2 + \left(1/\omega C\right)^2}}$$

Solving,

$$R^2 + \left(\frac{1}{\omega C}\right)^2 = \frac{4}{\omega^2 C^2}$$

or $R^2\omega^2C^2 = 3.$

Then,

$$\omega = 2\pi f = \frac{\sqrt{3}}{RC} \quad \rightarrow \quad f = \boxed{\frac{\sqrt{3}}{2\pi RC}}$$

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Additional Problems

P33.57 (a) We determine the number of turns from

$$\left(\Delta V_{2,\,\mathrm{rms}}\right) = \frac{N_2}{N_1}\left(\Delta V_{1,\,\mathrm{rms}}\right)$$

solving,

$$N_2 = \frac{(80)(2\,200\ \mathrm{V})}{110\ \mathrm{V}} = \boxed{1\,600\ \text{windings}}$$

(b) Assuming ideal conditions,

$$I_{1,\,\mathrm{rms}}\left(\Delta V_{1,\,\mathrm{rms}}\right) = I_{2,\,\mathrm{rms}}\left(\Delta V_{2,\,\mathrm{rms}}\right)$$

Solving for the rms current in the primary then gives

$$I_{1,\,\mathrm{rms}} = \frac{(1.50\ \mathrm{A})(2\,200\ \mathrm{V})}{110\ \mathrm{V}} = \boxed{30.0\ \mathrm{A}}$$

(c) For 95.0% efficiency,

$$0.950 I_{1,\,\mathrm{rms}}\left(\Delta V_{1,\,\mathrm{rms}}\right) = I_{2,\,\mathrm{rms}}\left(\Delta V_{2,\,\mathrm{rms}}\right)$$

and the rms current in the primary is

$$I_{1,\,\mathrm{rms}} = \frac{(1.20\ \mathrm{A})(2\,200\ \mathrm{V})}{(0.950)(110\ \mathrm{V})} = \boxed{25.3\ \mathrm{A}}$$

P33.58 From Equation 33.35, the resonance frequency for this circuit is

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi\sqrt{\left(2.80 \times 10^{-6}\ \mathrm{H}\right)\left(0.910 \times 10^{-12}\ \mathrm{F}\right)}} = 99.7\ \mathrm{MHz}$$

This frequency is not in the range of North American AM frequencies, which can be found from Internet research to be 520 kHz – 1 610 kHz. The frequency above is appropriate for an North American FM radio station.

P33.59 (a) The maximum voltage is given by

$$\Delta V_{\max} = \sqrt{\left(\Delta V_R\right)^2 + \left(\Delta V_L - \Delta V_C\right)^2}$$

$$= \sqrt{(20.0\ \mathrm{V})^2 + (25.0\ \mathrm{V} - 15.0\ \mathrm{V})^2}$$

$$= \boxed{22.4\ \mathrm{V}}$$

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(b) $\phi = \tan^{-1}\left(\dfrac{\Delta V_L - \Delta V_C}{\Delta V_R}\right) = \tan^{-1}\left(\dfrac{25.0\text{ V} - 15.0\text{ V}}{20.0\text{ V}}\right) = \boxed{26.6°}$

(c) The current amplitude I_{max} determines the voltage amplitude in each component of the circuit. We know the resistance R, so

$$\Delta V_R = I_{max}R \quad \rightarrow \quad I_{max} = \frac{\Delta V_R}{R} = \frac{20.0\text{ V}}{75.0\ \Omega} = \boxed{0.267\text{ A}}$$

(d) For the entire circuit,

$$\Delta V_{max} = I_{max}Z \quad \rightarrow \quad Z = \frac{\Delta V_{max}}{I_{max}} = \boxed{83.9\ \Omega}$$

(e) For the capacitor,

$$\Delta V_C = I_{max}X_C \quad \rightarrow \quad X_C = \frac{\Delta V_C}{I_{max}} = \frac{15.0\text{ V}}{I_{max}} = 56.3\ \Omega = \frac{1}{2\pi fC}$$

therefore,

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(60.0\text{ Hz})X_C} = 4.72\times10^{-5}\text{ F} = \boxed{47.2\ \mu\text{F}}$$

(f) For the inductor,

$$\Delta V_L = I_{max}X_L = I_{max}(2\pi fL)$$

therefore,

$$L = \frac{\Delta V_L}{2\pi f I_{max}} = \frac{25.0\text{ V}}{2\pi(60.0\text{ Hz})I_{max}} = \boxed{0.249\text{ H}}$$

(g) The average power delivered to the circuit is

$$P_{avg} = I_{rms}^2 R = \left(\frac{I_{max}}{\sqrt{2}}\right)^2 R = \boxed{2.67\text{ W}}$$

P33.60 We identify that $R = 200\ \Omega$, $L = 663$ mH, $C = 26.5\ \mu$F, $\omega = 377$ rad/s, and $\Delta V_{max} = 50.0$ V. So,

$$\omega L = 250\ \Omega, \text{ and } 1/\omega C = 100\ \Omega$$

The impedance is

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{(200\ \Omega)^2 + (250\ \Omega - 100\ \Omega)^2}$$
$$= 250\ \Omega$$

(a) The amplitude of the current is

$$I_{max} = \frac{\Delta V_{max}}{Z} = \frac{50.0\text{ V}}{250\ \Omega} = \boxed{0.200\text{ A}}$$

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The phase angle of the voltage relative to the current is

$$\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right) = \boxed{36.8^\circ}$$

with $\Delta v \nearrow$ leading $i \rightarrow$

(b) $\Delta V_{R,\max} = I_{\max} R = \boxed{40.0\text{ V}}$ at $\boxed{\phi = 0^\circ}$

(c) $\Delta V_{C,\max} = I_{\max} X_C = \boxed{20.0\text{ V}}$ at $\boxed{\phi = -90.0^\circ}$ (I leads ΔV)

(d) $\Delta V_{L,\max} = I_{\max} X_L = \boxed{50.0\text{ V}}$ at $\boxed{\phi = +90.0^\circ}$ (ΔV leads I)

P33.61 Consider a two-wire transmission line: each wire has resistance R, the total power transmitted is P, and the current in the wires is $I_{\text{rms}} = \dfrac{P}{\Delta V_{\text{rms}}}$. The power loss is 1.00% of the transmitted power P. Therefore,

$$P_{\text{loss}} = I_{\text{rms}}^2 R_{\text{line}} = \frac{P}{100}$$

$$P_{\text{loss}} = \left(\frac{P}{\Delta V_{\text{rms}}}\right)^2 (2R) = \frac{P}{100}$$

Solving for the resistance gives

$$R = \frac{\left(\Delta V_{\text{rms}}\right)^2}{200P}$$

The resistance of one wire is

$$R = \frac{\rho_{\text{Cu}}\ell}{A} = \frac{\left(\Delta V_{\text{rms}}\right)^2}{200P}$$

Solving for the area gives

$$A = \frac{\pi d^2}{4} = \frac{200\rho_{\text{Cu}} P \ell}{\left(\Delta V_{\text{rms}}\right)^2}$$

and the diameter is

$$d = \sqrt{\frac{800\rho_{\text{Cu}} P \ell}{\pi\left(\Delta V_{\text{rms}}\right)^2}}$$

$$= \sqrt{\frac{800(1.7\times10^{-8}\ \Omega\cdot\text{m})\ (20\,000\text{ W})(18\,000\text{ m})}{\pi\left(1.50\times10^{3}\text{ V}\right)^2}}$$

$$= 0.026\text{ m} = \boxed{2.6\text{ cm}}$$

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P33.62 Consider a two-wire transmission line: each wire has resistance R, the total power transmitted is P, and the current in the wires is $I_{\text{rms}} = \dfrac{P}{\Delta V_{\text{rms}}}$. The fractional power loss is f of the transmitted power P. Therefore,

$$P_{\text{loss}} = I_{\text{rms}}^2 R_{\text{line}} = fP$$

$$P_{\text{loss}} = \left(\frac{P}{\Delta V_{\text{rms}}}\right)^2 (2R) = fP \quad \rightarrow \quad R = \frac{f\left(\Delta V_{\text{rms}}\right)^2}{2P}$$

The resistance of one wire is

$$R = \frac{\rho_{\text{Cu}}\ell}{A} = \frac{\left(\Delta V_{\text{rms}}\right)^2}{200P}$$

Solving for the area gives

$$A = \frac{\pi d^2}{4} = \frac{200\rho_{\text{Cu}} P\ell}{\left(\Delta V_{\text{rms}}\right)^2}$$

and the diameter is

$$d = \boxed{\sqrt{\frac{8\rho_{\text{Cu}}\ell P}{\pi f\left(\Delta V_{\text{rms}}\right)^2}}}$$

P33.63 (a) The impedance is given by

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}$$

From which we obtain

$$X_C = X_L \pm \sqrt{Z^2 - R^2}$$

$$X_C = X_L + \sqrt{Z^2 - R^2}$$

$$= 700\ \Omega + \sqrt{(760\ \Omega)^2 - (400\ \Omega)^2} = 1\ 346\ \Omega = 1.35\ \text{k}\Omega$$

or

$$X_C = X_L - \sqrt{Z^2 - R^2}$$

$$= 700\ \Omega - \sqrt{(760\ \Omega)^2 - (400\ \Omega)^2} = 53.8\ \Omega$$

$\boxed{X_C \text{ could be } 53.8\ \Omega \text{ or it could be } 1.35\ \text{k}\Omega.}$

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(b) The power delivered to the circuit is given by

$$P = \left(I_{\text{rms}}\right)^2 R = \frac{\left(\Delta V_{\text{rms}}\right)^2}{Z^2} R$$

If the power is decreased as the frequency is raised, then the impedance is increased, so the inductive reactance is greater than the capacitive reactance, and the circuit must be above resonance:

$$X_L > X_C \quad \rightarrow \quad \omega L > 1/\omega C \quad \rightarrow \quad \omega > 1/\sqrt{LC} \quad \rightarrow \quad \omega > \omega_0$$

Therefore, the inductive reactance 700 Ω and the
capacitive reactance is 53.8 Ω.

(c) Now,

$$X_C = X_L \pm \sqrt{Z^2 - R^2} = 700 \pm \sqrt{(760)^2 - (200)^2} = 700 \pm 733$$

Here $X_C = 700 - 733 = -33\ \Omega$ is impossible, but
$X_C = 700 + 733 = 1433 = 1.43\ \text{k}\Omega$ is possible.

X_C must be 1.43 kΩ.

P33.64 The equation for $\Delta v(t)$ during the first period (using $y = mx + b$) is:

$$\Delta v = \frac{2\left(\Delta V_{\max}\right)t}{T} - \Delta V_{\max} = \Delta V_{\max}\left[\frac{2t}{T} - 1\right]$$

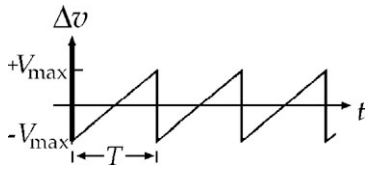
Therefore,

$$\left[(\Delta v)^2\right]_{\text{avg}} = \frac{1}{T}\int_0^T \left[\Delta v(t)\right]^2 dt = \frac{\left(\Delta V_{\max}\right)^2}{T}\int_0^T \left[\frac{2}{T}t - 1\right]^2 dt$$

$$\left[(\Delta v)^2\right]_{\text{avg}} = \frac{\left(\Delta V_{\max}\right)^2}{T}\left(\frac{T}{2}\right)\frac{\left[2t/T - 1\right]^3}{3}\Bigg|_{t=0}^{t=T}$$

$$= \frac{\left(\Delta V_{\max}\right)^2}{6}\left[(+1)^3 - (-1)^3\right] = \frac{\left(\Delta V_{\max}\right)^2}{3}$$

Then, $$\Delta V_{\text{rms}} = \sqrt{\left[(\Delta v)^2\right]_{\text{avg}}} = \sqrt{\frac{\left(\Delta V_{\max}\right)^2}{3}} = \boxed{\frac{\Delta V_{\max}}{\sqrt{3}}}$$



ANS. FIG. P33.64

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P33.65 The turns ratio is the factor of change in voltage:

$$\frac{N_1}{N_2} = \frac{\Delta V_1}{\Delta V_2}$$

with $Z_1 = \frac{\Delta V_1}{I_1}$ and $Z_2 = \frac{\Delta V_2}{I_2}$

we have $\frac{N_1}{N_2} = \frac{Z_1 I_1}{Z_2 I_2}$

(a) Since $\frac{I_1}{I_2} = \frac{N_2}{N_1}$, we find

$$\frac{N_1}{N_2} = \frac{Z_1 N_2}{Z_2 N_1} \quad \text{so} \quad \frac{N_1^2}{N_2^2} = \frac{Z_1}{Z_2} \quad \text{and} \quad \frac{N_1}{N_2} = \boxed{\sqrt{\frac{Z_1}{Z_2}}}$$

(b) $$\frac{N_1}{N_2} = \sqrt{\frac{8\,000\ \Omega}{8.00\ \Omega}} = \boxed{31.6}$$

P33.66 (a) The angular frequency is $\omega = 2\pi(60.0\text{ Hz}) = 377/\text{s}^{-1}$.

When *S* is open, *R*, *L*, and *C* are in series with the source, with impedance

$$Z = \frac{\Delta V_{\text{rms}}}{I_{\text{rms}}}$$

squaring both sides and substituting the definition of impedance gives

$$R^2 + (X_L - X_C)^2 = \left(\frac{\Delta V_{\text{rms}}}{I_{\text{rms}}}\right)^2 = \left(\frac{20.0\text{ V}}{0.183\text{ A}}\right)^2 = 1.194 \times 10^4\ \Omega^2 \quad \textbf{[1]}$$

When *S* is in position *a*, a parallel combination of two *R*'s presents equivalent resistance $\frac{R}{2}$, in series with *L* and *C*. The square of the impedance is then

$$\left(\frac{R}{2}\right)^2 + (X_L - X_C)^2 = \left(\frac{20.0\text{ V}}{0.298\text{ A}}\right)^2 = 4.504 \times 10^3\ \Omega^2 \quad \textbf{[2]}$$

When *S* is in position *b*, the current bypasses the inductor. *R* and *C* are in series with the source, and the square of the impedance is

$$R^2 + X_C^2 = \left(\frac{20.0\text{ V}}{0.137\text{ A}}\right)^2 = 2.131 \times 10^4\ \Omega^2 \quad \textbf{[3]}$$

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Subtract equation [2] from equation [1]:

$$\frac{3}{4}R^2 = 7.440\times10^3\ \Omega^2 \quad \rightarrow \quad \boxed{R = 99.6\ \Omega}$$

Only the positive root is physical, thus there is only one value for R.

(b) We have shown than only one resistance value is possible. Now equation [3] gives

$$X_C = \left[2.131\times10^4\ \Omega^2 - (99.6\ \Omega)^2\right]^{1/2} = 106.7\ \Omega = \frac{1}{\omega C}$$

Only the positive root is physical, thus there is only one value for C.

$$C = (\omega X_C)^{-1} = \left[(377\ \text{s}^{-1})106.7\ \Omega\right]^{-1}$$

$$= 2.49\times10^{-5}\ \text{F} = \boxed{24.9\ \mu\text{F}}$$

(c) Now equation [1] gives

$$X_L - X_C = \pm\left[1.194\times10^4\ \Omega^2 - (99.6\ \Omega)^2\right]^{1/2} = \pm 44.99\ \Omega$$

The equation shows us that there are two possible values for L.

$$X_L = X_C - 44.99\ \Omega = 106.7\ \Omega - 44.99\ \Omega\ = 61.74\ \Omega = \omega L$$

or

$$X_L = X_C + 44.99\ \Omega = 106.7\ \Omega + 44.99\ \Omega = 151.7\ \Omega\ = \omega L$$

then

$$L = \frac{X_L}{\omega} = 0.164\ \text{H or } 0.402\ \text{H} = \boxed{164\ \text{mH or } 402\ \text{mH}}$$

(d) From the calculations above, we see that only one value for R and only one value for C are possible. Two values for L are possible.

P33.67 (a) We can use $\sin A + \sin B = 2\sin\left(\frac{A}{2}+\frac{B}{2}\right)\cos\left(\frac{A}{2}-\frac{B}{2}\right)$ to find the sum of the two sine functions to be

$$E_1 + E_2 = (24.0\ \text{cm})\sin(4.50t + 35.0°)\cos 35.0°$$

$$E_1 + E_2 = (19.7\ \text{cm})\sin(4.50t + 35.0°)$$

Thus, the total wave has amplitude $\boxed{19.7\ \text{cm}}$ and has a constant phase difference of $\boxed{35.0°}$ from the first wave.

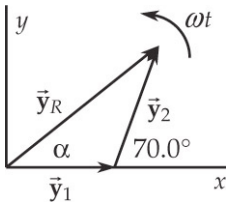
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(b) Refer to ANS. FIG. P33.67(b). In units of cm, the resultant phasor is

$$\vec{\mathbf{y}}_R = \vec{\mathbf{y}}_1 + \vec{\mathbf{y}}_2 = \left(12.0\hat{\mathbf{i}}\right) + \left(12.0\cos(70.0°)\hat{\mathbf{i}} + 12.0\sin(70.0)\hat{\mathbf{j}}\right)$$

$$= 16.1\hat{\mathbf{i}} + 11.3\hat{\mathbf{j}}$$

$$\vec{\mathbf{y}}_R = \sqrt{(16.1)^2 + (11.3)^2} \text{ at } \tan^{-1}\left(\frac{11.3}{16.1}\right) = \boxed{19.7 \text{ cm at } 35.0°}$$



ANS. FIG. P33.67(b)

(c) The answers are identical.

(d) Refer to ANS. FIG. P33.67(d). Adding the three waves yields

$$\vec{\mathbf{y}}_R = 12.0\cos(70.0°)\hat{\mathbf{i}} + 12.0\sin(70.0°)\hat{\mathbf{j}}$$

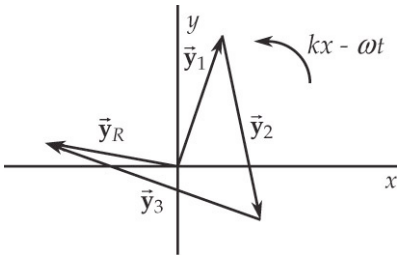
$$+ 15.5\cos(-80.0°)\hat{\mathbf{i}} + 15.5\sin(-80.0°)\hat{\mathbf{j}}$$

$$+ 17.0\cos(160°)\hat{\mathbf{i}} + 17.0\sin(160°)\hat{\mathbf{j}}$$

$$\vec{\mathbf{y}}_R = -9.18\hat{\mathbf{i}} + 1.83\hat{\mathbf{j}} = \boxed{9.36 \text{ cm at } 169°}$$

The wave function of the total wave is

$$y_R = (9.36 \text{ cm})\sin(15x - 4.5t + 169°)$$



ANS. FIG. P33.67(d)

P33.68 (a) Higher. At the resonance frequency, $X_L = X_C$. As the frequency increases, X_L goes up and X_C goes down.

(b) It is possible. We have three independent equations in the three unknowns L, C, and the certain f.

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(c) The equations are $\omega_0^2 = \frac{1}{LC} = 2\ 000\ \text{s}^{-1}$, $X_C = \frac{1}{\omega C} = 8.00\ \Omega$, and $X_L = \omega L = 12.0\ \Omega$. From the inductive reactance,

$$X_L = \omega L \quad \rightarrow \omega = \frac{X_L}{L}$$

then from the capacitive reactance,

$$X_C \omega = \frac{1}{\omega C}\frac{X_L}{L}$$

solving for the angular frequency gives

$$\omega^2 = \frac{X_L}{X_C}\frac{1}{LC} = \frac{X_L}{X_C}\omega_0^2 = \left(\frac{12.0\ \Omega}{8.00\ \Omega}\right)\left(2\ 000\ \text{s}^{-1}\right)^2$$

from which we obtain

$$\omega = 2\ 450\ \text{s}^{-1}$$

Then,

$$L = \frac{X_L}{\omega} = \frac{12.0\ \Omega}{2\ 450\ \text{s}^{-1}} = 4.90 \times 10^{-3}\ \text{H} = \boxed{4.90\ \text{mH}}$$

$$C = \frac{1}{\omega X_C} = \frac{1}{\left(2\ 450\ \text{s}^{-1}\right)(8.00\ \Omega)} = 5.10 \times 10^{-5}\ \text{F} = \boxed{51.0\ \mu\text{F}}$$

P33.69 (a) The lowest-frequency standing-wave pattern is N-A-N. The distance between the clamps we represent as $d = d_{\text{NN}} = \frac{\lambda}{2}$. The speed of transverse waves on the string is $v = f\lambda = \sqrt{\frac{T}{\mu}} = f2d$.

The magnetic force on the wire oscillates at 60 Hz, so the wire will oscillate in resonance at 60 Hz. From the speed of transverse waves,

$$v = f\lambda = \sqrt{\frac{T}{\mu}} = f2d$$

we obtain the period as

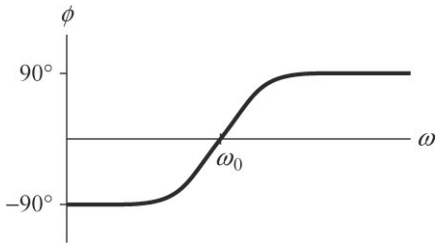
$$T = 4\mu f^2 d^2 = 4\left(19.0 \times 10^{-3}\ \text{kg/m}\right)(60.0\ \text{Hz})^2 d^2$$
$$= \left(274\ \text{kg/m}\cdot\text{s}^2\right)d^2$$

Tension T and separation d must be related by $T = 274d^2$ where T is in newtons and d is in meters.

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(b) One possibility is $T = 10.9$ N and $d = 0.200$ m. Any values of T and d related according to this expression will work. We did not need to use the value of the current and magnetic field.

P33.70 (a) See the graph in ANS. FIG. P33.70(a).



ANS. FIG. P33.70(a)

(b) $\phi = \tan^{-1}\left(\dfrac{\omega L - 1/\omega C}{R}\right)$ changes from –90° for $\omega = 0$ to 0 at the resonance frequency to +90° as ω goes to infinity.

The slope of the graph is $d\phi/d\omega$:

$$\frac{d\phi}{d\omega} = \frac{1}{1+\left(\dfrac{\omega L - 1/\omega C}{R}\right)^2}\frac{1}{R}\left(L - \frac{1}{C}(-1)\frac{1}{\omega^2}\right)$$

$$= \boxed{\frac{R}{R^2 + (\omega L - 1/\omega C)^2}\left(L + \frac{1}{\omega^2 C}\right)}$$

(c) At resonance we have $\omega_0^2 = 1/LC$; substituting, we find the slope at the resonance point is

$$\left.\frac{d\phi}{d\omega}\right|_{\omega_0} = \frac{1}{R+0^2}\left(L + \frac{LC}{C}\right) = \frac{2L}{R} = \frac{2Q}{\omega_0}$$

where $Q = \omega_0 L/R$.

P33.71 (a) When ωL is very large, Z is large for the bottom branch, so it carries negligible current. Also, $\dfrac{1}{\omega C}$ will be negligible compared to R for the top branch, so

$$I = \frac{V}{Z} \approx \frac{V}{R} = \frac{45.0\text{ V}}{200\ \Omega} = \boxed{0.225\text{ A}}$$

flows in the power supply and the top branch.

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(b) Now $\frac{1}{\omega C}$ is very large in the top branch and ωL is very small compared to R in the bottom branch; the generator and bottom branch carry

$$I = \frac{V}{Z} \approx \frac{V}{R} = \frac{45.0\text{ V}}{100\ \Omega} = \boxed{0.450\text{ A}}$$

P33.72 (a) With both switches closed, the current goes only through the generator and resistor.

$$\boxed{i = \frac{\Delta V_{\max}}{R}\cos\omega t}$$

(b) $$\boxed{P = \frac{1}{2}\frac{\left(\Delta V_{\max}\right)^2}{R}}$$

(c) $$\boxed{i = \frac{\Delta V_{\max}}{\sqrt{R^2 + \omega^2 L^2}}\cos\left[\omega t + \tan^{-1}\left(\frac{\omega L}{R}\right)\right]}$$

(d) For $0 = \phi = \tan^{-1}\left(\frac{\omega_0 L - \left(1/\omega_0 C\right)}{R}\right)$,

We require $\omega_0 L = \frac{1}{\omega_0 C}$, so $\boxed{C = \frac{1}{\omega_0^2 L}}$

(e) The frequency is the resonance frequency: $Z = \boxed{R}$

For parts (f) and (g), the circuit is at resonance, so $Z = R$ and $X_C = X_L = \omega_0 L$.

(f) To find the maximum energy stored in the capacitor, we start with

$$U = \frac{1}{2}C\left(\Delta V_C\right)^2 = \frac{1}{2}C\left(IX_C\right)^2$$

When $I = I_{\max}$,

$$U_{\max} = \frac{1}{2}CI_{\max}^2 X_C^2 = \frac{1}{2}C\left(\frac{\Delta V_{\max}}{R}\right)^2\left(\omega_0 L\right)^2 = \boxed{\frac{\left(\Delta V_{\max}\right)^2 L}{2R^2}}$$

(g) $$U_{\max} = \frac{1}{2}LI_{\max}^2 = \boxed{\frac{1}{2}L\frac{\left(\Delta V_{\max}\right)^2}{R^2}}$$

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(h) Now $\omega = 2\omega_0 = \dfrac{2}{\sqrt{LC}}$, so

$$\phi = \tan^{-1}\left(\frac{\omega L - (1/\omega C)}{R}\right) = \tan^{-1}\left(\frac{2\sqrt{L/C} - (1/2)\sqrt{L/C}}{R}\right)$$

$$= \boxed{\tan^{-1}\left(\frac{3}{2R}\sqrt{\frac{L}{C}}\right)}$$

(i) Now $\omega L = \dfrac{1}{2}\dfrac{1}{\omega C}$, so $\omega = \boxed{\dfrac{1}{\sqrt{2LC}}} = \dfrac{\omega_0}{\sqrt{2}}$

P33.73 (a) The inductive reactance of the circuit is

$$X_L = 2\pi f L = 2\pi(50.0\text{ Hz})(0.250\text{ H}) = \boxed{78.5\ \Omega}$$

(b) The capacitive reactance of the circuit is

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(50.0\text{ Hz})(2.00\times10^{-6}\text{ F})}$$

$$= 1.59\times10^{3}\ \Omega = \boxed{1.59\ \text{k}\Omega}$$

(c) The impedance of the circuit is

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(150\ \Omega)^2 + (78.5\ \Omega - 1590\ \Omega)^2}$$

$$= 1.52\times10^{3}\ \Omega = \boxed{1.52\ \text{k}\Omega}$$

(d) The maximum current is

$$I_{\max} = \frac{\Delta V_{\max}}{Z} = \frac{2.10\times10^{2}\text{ V}}{1.52\times10^{3}\ \Omega} = 0.138\text{ A} = \boxed{138\text{ mA}}$$

(e) $\phi = \tan^{-1}\left(\dfrac{X_L - X_C}{R}\right) = \tan^{-1}\left(\dfrac{78.5\ \Omega - 1590\ \Omega}{150\ \Omega}\right) = \boxed{-84.3°}$

(f) $\cos\phi = \cos\left[\tan^{-1}\left(\dfrac{78.5\ \Omega - 1\ 590\ \Omega}{150\ \Omega}\right)\right] = \boxed{0.098\ 7}$

(g) The power input into the circuit is

$$P = I_{\text{rms}}\Delta V_{\text{rms}}\cos\phi = \frac{(\Delta V_{\text{rms}})^2}{Z}\cos\phi = \frac{(\Delta V_{\max}/\sqrt{2})^2}{Z}\cos\phi$$

$$= \frac{(\Delta V_{\max})^2}{2Z}\cos\phi$$

$$P = \frac{(210\text{ V})^2}{2(1.52\times10^{3}\ \Omega)}(0.098\ 7) = \boxed{1.43\text{ W}}$$

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P33.74 (a) We are given $X_L = X_C = 1\,884\ \Omega$ when $f = 2\,000$ Hz. The impedance is then

$$L = \frac{X_L}{2\pi f} = \frac{1\,884\ \Omega}{4\,000\pi\ \text{rad/s}} = 0.150\ \text{H}$$

and the capacitance is

$$C = \frac{1}{(2\pi f)X_C} = \frac{1}{(4\,000\pi\ \text{rad/s})(1\,884\ \Omega)}$$
$$= 42.2\ \text{nF}$$

therefore,

$$X_L = 2\pi f(0.150\ \text{H})$$

$$X_C = \frac{1}{(2\pi f)(4.22\times 10^{-8}\ \text{F})}$$

and

$$Z = \sqrt{(40.0\ \Omega)^2 + (X_L - X_C)^2}$$

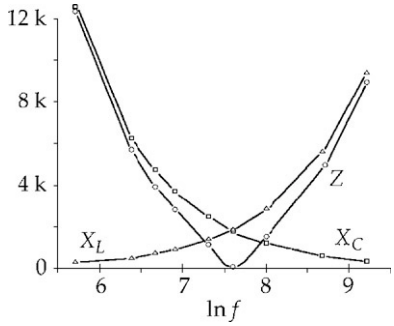
TABLE P33.74 lists the inductive reactance, the capacitive reactance, and the impedance for the frequencies listed in the problem statement.

f (Hz)	X_L (Ω)	X_C (Ω)	Z (Ω)
300	283	12 600	12 300
600	565	6 280	5 720
800	754	4 710	3 960
1 000	942	3 770	2 830
1 500	1 410	2 510	1 100
2 000	1 880	1 880	40
3 000	2 830	1 260	1 570
4 000	3 770	942	2 830
6 000	5 650	628	5 020
10 000	9 420	377	9 040

TABLE P33.74

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(b) ANS. FIG. P33.74(b) shows a graph of X_L, X_C, and Z as a function of the frequency f.



ANS. FIG. P33.74(b)

P33.75 The resonance frequency is

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.050\,0\text{ H})(5.00\times10^{-6}\text{ F})}} = 318\text{ Hz}$$

The operating frequency is $f = f_0/2 = 159$ Hz. We can calculate the impedance at this frequency. The inductive reactance is

$$X_L = 2\pi f L = 2\pi(159\text{ Hz})(0.050\,0\text{ H}) = 50.0\ \Omega$$

The capacitive reactance is

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(159\text{ Hz})(5.00\times10^{-6}\text{ F})} = 200\ \Omega$$

The impedance is

$$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2} = \sqrt{8.00^2 + (50.0 - 200)^2}\ \Omega = 150\ \Omega$$

So the current is

$$I_{\text{rms}} = \frac{\Delta V_{\text{rms}}}{Z} = \frac{400\ V}{150\ \Omega} = 2.66\text{ A}$$

The power delivered by the source is the power delivered to the resistor:

$$P = (2.66\text{ A})^2\left(8.00\Omega\right) = \boxed{56.7\text{ W}}$$

P33.76 (a) At resonance,

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\left(1.00\times10^{-3}\text{ H}\right)\left(1.00\times10^{-9}\text{ F}\right)}} = 1.00\times10^{6}\text{ rad/s}$$

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At that point,

$$Z = R = 1.00\ \Omega \quad \text{and} \quad I = \frac{1.00\ \text{V}}{1.00\ \Omega} = 1.00\ \text{A}$$

The power is

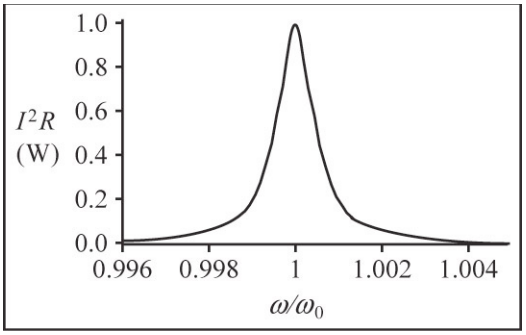
$$P = I^2R = (1.00\ \text{A})^2(1.00\ \Omega) = 1.00\ \text{W}$$

We compute the power at some other angular frequencies, listed in TABLE P33.76.

$\frac{\omega}{\omega_0}$	$\omega L\ (\Omega)$	$\frac{1}{\omega C}\ (\Omega)$	$Z\ (\Omega)$	$P = I_{\text{rms}}^{\ 2}\ R\ (W)$
0.9990	999.0	1001.0	2.24	0.19984
0.9991	999.1	1000.9	2.06	0.23569
0.9993	999.3	1000.7	1.72	0.33768
0.9995	999.5	1000.5	1.41	0.49987
0.9997	999.7	1000.3	1.17	0.73524
0.9999	999.9	1000.1	1.02	0.96153
1.0000	1000	1000.0	1.00	1.00000
1.0001	1000.1	999.9	1.02	0.96154
1.0003	1000.3	999.7	1.17	0.73535
1.0005	1000.5	999.5	1.41	0.50012
1.0007	1000.7	999.3	1.72	0.33799
1.0009	1000.9	999.1	2.06	0.23601
1.0010	1001	999.0	2.24	0.20016

TABLE P33.76

ANS. FIG. P33.76 shows a graph of the results tabulated above.



ANS. FIG. P33.76

(b) The angular frequencies giving half the maximum power are

$$0.999\,5 \times 10^6\ \text{rad/s} \quad \text{and} \quad 1.000\,5 \times 10^6\ \text{rad/s}$$

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so the full width at half the maximum is

$$\Delta\omega = (1.000\,5 - 0.999\,5) \times 10^6 \text{ rad/s}$$

$$\Delta\omega = 1.00 \times 10^3 \text{ rad/s}$$

Since $\Delta\omega = 2\pi\,\Delta f$, $\quad \Delta f = 159$ Hz

and for comparison,

$$\frac{R}{2\pi L} = \frac{1.00\ \Omega}{2\pi\left(1.00 \times 10^{-3}\text{H}\right)} = 159 \text{ Hz}$$

The two quantities agree.

Challenge Problems

P33.77 We start with

$$\frac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}} = \frac{IR}{IZ} = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + \left(X_L - X_C\right)^2}}$$

$$= \frac{8.00\ \Omega}{\sqrt{(8.00\ \Omega)^2 + \left[2\pi fL - 1/2\pi fC\right]^2}}$$

Then, at 200 Hz,

$$\left(\frac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}}\right)^2 = \left(\frac{R}{Z}\right)^2 = \frac{1}{4} = \frac{(8.00\ \Omega)^2}{(8.00\ \Omega)^2 + \left[400\pi L - 1/400\pi C\right]^2}$$

and at 4 000 Hz,

$$\left(\frac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}}\right)^2 = \left(\frac{R}{Z}\right)^2 = \frac{1}{4} = \frac{(8.00\ \Omega)^2}{(8.00\ \Omega)^2 + \left[8000\pi L - 1/8000\pi C\right]^2}$$

At the low frequency, $X_L - X_C < 0$. This reduces to

$$400\pi L - \frac{1}{400\pi C} = -13.9\ \Omega \qquad \textbf{[1]}$$

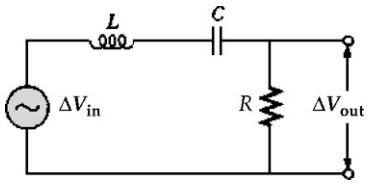
For the high frequency half-voltage point,

$$8\,000\pi L - \frac{1}{8000\pi C} = +13.9\ \Omega \qquad \textbf{[2]}$$

Solving equations [1] and [2] simultaneously gives

(a) $L = \boxed{580\ \mu\text{H}}$

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ANS. FIG. P33.77(a)

(b) $C = \boxed{54.6\ \mu\text{F}}$

(c) When $X_L = X_C$, $\dfrac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}} = \left(\dfrac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}}\right)_{\max} = \boxed{1.00}$

(d) $X_L = X_C$ requires

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{\left(5.80\times10^{-4}\ \text{H}\right)\left(5.46\times10^{-5}\ \text{F}\right)}} = \boxed{894\ \text{Hz}}$$

(e) $\boxed{\text{At 200 Hz}}$, $\dfrac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}} = \dfrac{R}{Z} = \dfrac{1}{2}$ and $X_C > X_L$, so the phasor diagram is as shown in ANS. FIG. P33.77(e).

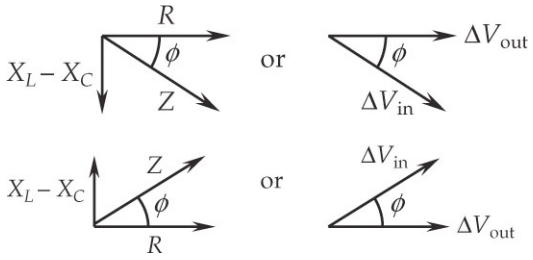
$$\boxed{\phi =} -\cos^{-1}\left(\frac{R}{Z}\right) = -\cos^{-1}\left(\frac{1}{2}\right) = \boxed{-60.0^\circ}$$

so $\boxed{\left(\Delta v_{\text{out}}\ \text{leads}\ \Delta v_{\text{in}}\right)}$.

$\boxed{\text{At } f_0}$, $X_L = X_C$ so

$$\boxed{\phi = 0\ \left(\Delta v_{\text{out}}\ \text{is in phase with}\ \Delta v_{\text{in}}\right)}.$$

$\boxed{\text{At 4 000 Hz}}$, $\dfrac{\Delta v_{\text{out}}}{\Delta v_{\text{in}}} = \dfrac{R}{Z} = \dfrac{1}{2}$ and $X_L - X_C > 0$.



ANS. FIG. P33.77(e)

Thus, $\boxed{\phi =} \cos^{-1}\left(\dfrac{1}{2}\right) = \boxed{+60.0^\circ}$

or $\boxed{\Delta v_{\text{out}}\ \text{lags}\ \Delta v_{\text{in}}}$.

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(f) At 200 Hz and at 4 kHz,

$$P = \frac{\left(\Delta v_{\text{out, rms}}\right)^2}{R} = \frac{\left[(1/2)\Delta v_{\text{in, rms}}\right]^2}{R} = \frac{(1/4)\left(\Delta v_{\text{in, max}}/\sqrt{2}\right)^2}{R}$$

$$= \frac{(1/2)(10.0\text{ V})^2}{4(8.00\ \Omega)} = \boxed{1.56\text{ W}}$$

At f_0,

$$P = \frac{\left(\Delta v_{\text{out, rms}}\right)^2}{R} = \frac{\left(\Delta v_{\text{in, rms}}\right)^2}{R} = \frac{\left(\Delta v_{\text{in, max}}/\sqrt{2}\right)^2}{R} = \frac{(1/2)(10.0\text{ V})^2}{(8.00\ \Omega)}$$

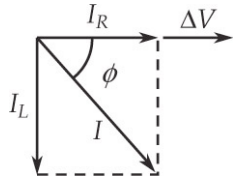
$$= \boxed{6.25\text{ W}}$$

(g) We take

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi f_0 L}{R} = \frac{2\pi(894\text{ Hz})\left(5.80\times10^{-4}\text{ H}\right)}{8.00\ \Omega} = \boxed{0.408}$$

P33.78 (a) $I_{R,\text{ rms}} = \dfrac{\Delta V_{\text{rms}}}{R} = \dfrac{100\text{ V}}{80.0\ \Omega} = \boxed{1.25\text{ A}}$

(b) The total current will $\boxed{\text{lag}}$ the applied voltage as seen in the phasor diagram shown in ANS. FIG. P33.78.



ANS. FIG. P33.78

$$I_{L,\text{ rms}} = \frac{\Delta V_{\text{rms}}}{X_L}$$

$$= \frac{100\text{ V}}{2\pi\left(60.0\text{ s}^{-1}\right)(0.200\text{ H})} = 1.33\text{ A}$$

Thus, the phase angle is:

$$\phi = \tan^{-1}\left(\frac{I_{L,\text{ rms}}}{I_{R,\text{ rms}}}\right) = \tan^{-1}\left(\frac{1.33\text{ A}}{1.25\text{ A}}\right) = \boxed{46.7^\circ}$$

P33.79 We are given $L = 2.00$ H, $C = 10.0 \times 10^{-6}$ F, $R = 10.0\ \Omega$, and $\Delta v = (100\sin\omega t)$.

(a) The resonance frequency ω_0 produces the maximum current and thus the maximum power delivery to the resistor.

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(2.00\text{ H})\left(10.0\times10^{-6}\text{ F}\right)}} = \boxed{224\text{ s}^{-1}}$$

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(b) At the resonance frequency ω_0, the impedance $Z = R$, and

$$P = I_{\text{rms}}^2 R$$

$$= \left(\frac{\Delta V_{\text{rms}}}{Z}\right)^2 R = \frac{\left(\Delta V_{\text{rms}}\right)^2}{Z^2} R = \frac{\left(\Delta V_{\text{rms}}\right)^2}{R^2} R = \frac{\left(\Delta V_{\text{max}}/\sqrt{2}\right)^2}{R}$$

$$= \frac{(100\ \text{V})^2}{2(10.0\ \Omega)} = \boxed{500\ \text{W}}$$

(c) Now,

$$P = \frac{1}{2} P_{\text{max}} \quad \rightarrow \quad \frac{\left(\Delta V_{\text{rms}}\right)^2}{Z^2} R = \frac{1}{2} \frac{\left(\Delta V_{\text{rms}}\right)^2}{R}$$

So, $Z^2 = 2R^2$, or

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

which is a fourth order equation in ω. But this can be simplified to two equations:

$$\omega L - \frac{1}{\omega C} = \pm R \quad \rightarrow \quad \omega^2 LC \mp \omega CR - 1 = 0$$

The angular frequency ω must be positive, so we solve for the positive roots. (In the following, we suppress all units.)

For $\omega^2 LC - \omega CR - 1 = 0$,

$$\omega = \frac{-\left(-CR\right) + \sqrt{(-CR)^2 - 4LC(-1)}}{2LC}$$

$$= \frac{R + \sqrt{R^2 + 4L/C}}{2L}$$

$$= \frac{10.0 + \sqrt{(10.0)^2 + 4(2.00)/\left(10.0 \times 10^{-6}\right)}}{2(2.00)} = \boxed{226\ \text{s}^{-1}}$$

For $\omega^2 LC + \omega CR - 1 = 0$,

$$\omega = \frac{-(CR) + \sqrt{(-CR)^2 - 4LC(-1)}}{2LC}$$

$$= \frac{-R + \sqrt{R^2 + 4L/C}}{2L}$$

$$= \frac{-10.0 + \sqrt{(10.0)^2 + 4(2.00)/\left(10.0 \times 10^{-6}\right)}}{2(2.00)} = \boxed{221\ \text{s}^{-1}}$$

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P33.80 The currents in the three components of the circuit are

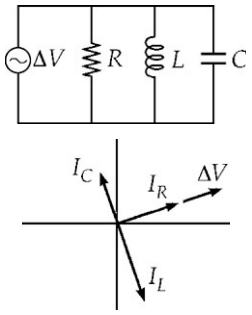
$$I_R = \frac{\Delta V_{\text{rms}}}{R},\ I_L = \frac{\Delta V_{\text{rms}}}{X_L} = \frac{\Delta V_{\text{rms}}}{\omega L},\ \text{and}\ I_C = \frac{\Delta V_{\text{rms}}}{X_C} = \frac{\Delta V_{\text{rms}}}{(\omega C)^{-1}}$$

(a) Then,

$$I_{\text{rms}} = \left[I_R^2 + (I_C - I_L)^2 \right]^{1/2} = \boxed{\Delta V_{\text{rms}} \left[\left(\frac{1}{R^2} \right) + \left(\omega C - \frac{1}{\omega L} \right)^2 \right]^{1/2}}$$

(b) $$\tan\phi = \frac{I_C - I_L}{I_R} = \Delta V_{\text{rms}} \left[\frac{1}{X_C} - \frac{1}{X_L} \right] \left(\frac{1}{\Delta V_{\text{rms}}/R} \right)$$

$$\boxed{\tan\phi = R\left(\frac{1}{X_C} - \frac{1}{X_L} \right)}$$



ANS. FIG. P33.80

P33.81 We have $P = I_{\text{rms}}^2 R = \left(\frac{\Delta V_{\text{rms}}}{Z} \right)^2 R = \frac{(\Delta V_{\text{rms}})^2}{Z^2} R$, and

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

Therefore,

$$Z^2 = \frac{(\Delta V_{\text{rms}})^2 R}{P} \qquad \rightarrow \qquad R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 = \frac{(\Delta V_{\text{rms}})^2 R}{P}$$

$$\left(\omega L - \frac{1}{\omega C} \right)^2 = \frac{(\Delta V_{\text{rms}})^2 R}{P} - R^2$$

which is a fourth order equation in ω. But this can be simplified to two equations:

$$\omega L - \frac{1}{\omega C} = \pm A \qquad \rightarrow \qquad \omega^2 LC \mp \omega CA - 1 = 0$$

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where $A = \sqrt{\dfrac{(\Delta V_{\text{rms}})^2 R}{P} - R^2}$.

We will solve for ω when ΔV_{rms} = 100 V and P = 250 W. From Figure P33.24, we have R = 40.0 Ω, L = 185 mH = 0.185 H, and C = 65.0 μF = 65.0 × 10^{-6} F.

The quantity A is

$$A = \sqrt{\frac{(\Delta V_{\text{rms}})^2 R}{P} - R^2} = \sqrt{\frac{(120\ \text{V})^2 (40.0\ \Omega)}{250\ \text{W}} - (40.0\ \Omega)^2}$$
$$= \sqrt{704}\ \Omega$$

The angular frequency ω must be positive, so we solve for the positive roots. (In the following, we suppress all units.)

For $\omega^2 LC - \omega CA - 1 = 0$,

$$\omega = \frac{-(-CA) + \sqrt{(-CA)^2 - 4LC(-1)}}{2LC}$$
$$= \frac{A + \sqrt{A^2 + 4L/C}}{2L}$$
$$= \frac{\sqrt{704} + \sqrt{704 + 4(0.185)/\left(65.0 \times 10^{-6}\right)}}{2(0.185)}$$
$$= 226\ \text{s}^{-1} = 2\pi f \quad \rightarrow \quad f = \boxed{58.7\ \text{Hz}}$$

For $\omega^2 LC + \omega CA - 1 = 0$,

$$\omega = \frac{-(CA) + \sqrt{(-CA)^2 - 4LC(-1)}}{2LC}$$
$$= \frac{-A + \sqrt{A^2 + 4L/C}}{2L}$$
$$= \frac{-\sqrt{704} + \sqrt{704 + 4(0.185)/\left(65.0 \times 10^{-6}\right)}}{2(0.185)}$$
$$= 225\ \text{s}^{-1} = 2\pi f \quad \rightarrow \quad f = \boxed{35.9\ \text{Hz}}$$

There are two answers because the circuit can be either above or below resonance.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P33.2 (a) 193 Ω; (b) 144 Ω

P33.4 (a) 170 V; (b) 2.40×10^2 Ω;
(c) Because $P_{\text{avg}} = \dfrac{(\Delta V_{\text{rms}})^2}{R} \quad \rightarrow \quad R = \dfrac{(\Delta V_{\text{rms}})^2}{P_{\text{avg}}}$, a 100-W bulb has less resistance than a 60.0-W bulb.

P33.6 (a) 25.3 rad/s; (b) 0.114 s

P33.8 (a) The rms current in each 150-W bulb is 1.25 A. The rms current in the 100-W bulb is 0.833 A; (b) $R_1 = 96.0\ \Omega,\ R_2 = 96.0\ \Omega,$ and $R_3 = 144\ \Omega$; (c) 36.0 Ω

P33.10 (a) 0.0424 H; (b) 942 rad/s

P33.12 0.750 H

P33.14 3.80 J

P33.16 (a) 15.0 Hz; (b) 84.9 V; (c) 47.1 Ω; (d) 1.80 A; (e) 2.55 A

P33.18 (a) 221 Ω; (b) 0.163 A; (c) 0.230 A; (d) no

P33.20 (a) 69.3 V; (b) 40.0 Hz; (c) 20.3 μF

P33.22 $\sqrt{2}C(\Delta V_{\text{rms}})$

P33.24 (a) 146 V ; (b) 212 V; (c) 179 V; (d) 33.4 V

P33.26 (a) 109 Ω; (b) $I_{\max} = 0.367$ A; (c) $\omega = 100$ rad/s;
(d) $\phi = -0.896$ rad $= -51.3°$

P33.28 2.79 kHz

P33.30 See ANS. FIG P33.30.

P33.32 (a) 88.4 Ω; (b) 107 Ω; (c) 1.12 A; (d) the voltage lags behind the current by 55.8°; (e) Adding an inductor will change the impedence, and hence the current in the circuit. The current could be larger or smaller, depending on the inductance added. The largest current would result when the inductive reactance equals the capacitive reactance, the impedance has its minimum value, equal to 60.0 Ω, and the current in the circuit is

$$I_{\max} = \frac{\Delta V_{\max}}{Z} = \frac{\Delta V_{\max}}{R} = \frac{1.20 \times 10^2\ \text{V}}{60.0\ \Omega} = 2.00\ \text{A}$$

P33.34 In order for the power factor to be equal to 1.00, we would have to have $X_L = 0$, which would require either L or f to be zero. Because this

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is not the case, the situation is impossible.

P33.36 8.00 W

P33.38 (a) 66.8 Ω; (b) 0.953 A; (c) 45.4 W

P33.40 (a) 5.43 A; (b) 0.905; (c) 281 μF; (d) 109 V

P33.42 (a) 3.56 kHz; (b) 5.00 A; (c) 22.4; (d) 2.24 kV

P33.44 (a) 156 pH; (b) 8.84 Ω

P33.46 $\dfrac{4\pi RC\sqrt{LC}\left(\Delta V_{\text{rms}}\right)^2}{4R^2C+9L}$

P33.48 (a) 9.23 V; (b) 2.40 W

P33.50 (a) 29.0 kW; (b) 5.80×10^{-3}; (c) It is impossible to transmit so much power at such low voltage.

P33.52 1.88 V

P33.54 (a) $\dfrac{R}{\sqrt{R^2+\left(1/\omega C\right)^2}}$; (b) 0; (c) 1

P33.56 (a) $\dfrac{1/\omega C}{\sqrt{R^2+\left(1/\omega C\right)^2}}$; (b) 1; (c) 0; (d) $\dfrac{\sqrt{3}}{2\pi RC}$

P33.58 The resonance frequency for this circuit is not in the North American AM frequency range.

P33.60 (a) 0.200 A, 36.8°; (b) 40.0 V at $\phi = 0°$; (c) 20.0 V at $\phi = -90.0°$; (d) 50.0 V at $\phi = +90.0°$

P33.62 $\sqrt{\dfrac{8\rho_{\text{Cu}}\,\ell P}{\pi f\left(\Delta V_{\text{rms}}\right)^2}}$

P33.64 See P33.64 for full explanation.

P33.66 (a) R = 99.6 Ω; (b) 24.9 μF; (c) 164 mH or 402 mH; (d) Only one value for R and only one value for C are possible. Two values for L are possible.

P33.68 (a) Higher. At the resonance frequency, $X_L = X_C$. As the frequency increases, X_L goes up and X_C goes down; (b) It is possible. We have three independent equations in the three unknowns L, C, and the certain f; (c) L = 4.90 mH and C = 51.0 μF

P33.70 (a) See ANS. FIG. P33.70(a); (b) $\dfrac{R}{R^2+\left(\omega L-1/\omega C\right)^2}\left(L+\dfrac{1}{\omega^2 C}\right)$; (c) See P33.70(c) for full explanation.

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P33.72 (a) $i = \dfrac{\Delta V_{\max}}{R}\cos\omega t$; (b) $P = \dfrac{1}{2}\dfrac{(\Delta V_{\max})^2}{R}$;

(c) $i = \dfrac{\Delta V_{\max}}{\sqrt{R^2+\omega^2 L^2}}\cos\left[\omega t + \tan^{-1}\left(\dfrac{\omega L}{R}\right)\right]$; (d) $C = \dfrac{1}{\omega_0^2 L}$; (e) R;

(f) $\dfrac{(\Delta V_{\max})^2 L}{2R^2}$; (g) $\dfrac{1}{2}L\dfrac{(\Delta V_{\max})^2}{R^2}$; (h) $\tan^{-1}\left(\dfrac{3}{2R}\sqrt{\dfrac{L}{C}}\right)$; (i) $\dfrac{1}{\sqrt{2LC}}$

P33.74 (a) See Table P33.74; (b) See ANS. FIG. P33.74(b).

P33.76 (a) See ANS. FIG. P33.76; (b) See P33.76 for full explanation.

P33.78 (a) 1.25 A; (b) lag, 46.7°

P33.80 (a) $\Delta V_{\text{rms}}\left[\left(\dfrac{1}{R^2}\right)+\left(\omega C - \dfrac{1}{\omega L}\right)^2\right]^{1/2}$; (b) $\tan\phi = R\left(\dfrac{1}{X_C}-\dfrac{1}{X_L}\right)$

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